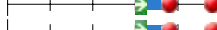
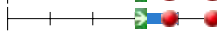
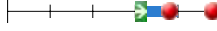


Program overview

14-Dec-2022 13:25

Year 2022/2023
Organization Civil Engineering and Geosciences
Education Master Environmental Engineering

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
ENVE Master 2022			
ENVE Year 1			
ENVE Faculty Core Module (MUDE)			
CEGM1000	Modelling, Uncertainty and Data for Engineers	12	
ENVE Programme: Principles of Environmental Engineering			
ENVM1000	Principles of Environmental Engineering	9	
ENVE Track Atmospheric Environment Engineering (AEE)			
ENVM1700	Fundamentals of AEE	15	
ENVM1800	Atmospheric Measurements and Modelling	9	
ENVM1900	Grand Challenges in AEE	15	
ENVE Track Water Resources Engineering (WRE)			
ENVM1400	Quality, Quantity and Design, Information and Advice	15	
ENVM1500	Water Quality and Principles of Environmental Engineering	9	
ENVM1501	Design and Modeling of Urban Water Infrastructure Systems	9	
ENVM1502	River Basin Hydrology and Water Management	9	
ENVM1600	Water Treatment Technologies	15	
ENVM1601	Operation, Control, Management and Adoption of Urban Water Infrastructure Systems	15	
ENVM1602	Regional Hydrology	15	
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ENVE Track Resource & Waste Engineering (R&WE)			
ENVM1100	Fundamentals of RW and E	15	
ENVM1200	Resource Engineering	9	
ENVM1300	Waste Processing Technologies	15	
ENVM1301	Reactive Resources and Wastes	15	

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE Master 2022

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE Year 1

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE Faculty Core Module (MUDE)
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CEGM1000	Modelling, Uncertainty and Data for Engineers	12
Responsible Instructor	Dr.ir. A.A. Verhagen	
Module Manager	Dr.ir. A.A. Verhagen	
Instructor	Dr.ir. W. Daamen	
Instructor	Dr.ir. E.C. Slob	
Instructor	Dr.ir. C.C.J.M. Tiberius	
Instructor	Dr. A. Amiri Simkooei	
Instructor	F.P. van der Meer	
Instructor	Dr.ir. H.R. Schipper	
Instructor	Dr. A.A. Nunez Vicencio	
Instructor	A. Tsouvalas	
Instructor	Dr.ir. O. Morales Napoles	
Instructor	Dr.ir. R.W. Hut	
Instructor	Dr. V.L. Knoop	
Instructor	Dr.ir. G. Homem de Almeida Correia	
Instructor	Dr. M.F.S. Tissier	
Instructor	Dr.ir. I. Barcelos Carneiro M Da R	
Instructor	P.K. Krishnakumari	
Instructor	Dr. A. Cabboi	
Instructor	Dr.ir. A.P. van den Eijnden	
Instructor	Dr. D.V. Voskov	
Instructor	Dr. H. Wang	
Instructor	Dr.ir. M. Nogal Macho	
Instructor	Dr.ir. R.E.F. Lindeboom	
Instructor	Dr.ir. R.C. Lanzafame	
Instructor	E. Ragno	
Instructor	Dr. R. Taormina	
Instructor	Dr. A. Cicirello	
Instructor	J.O. Colomes Gene	
Instructor	M. Rinaldi	
Instructor	A. Jagadeesh	
Instructor	Dr. P.M.N. Mares Nasarre	
Contact Hours / Week x/x/x/x	4HC, 4W, 4PJ	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	All students have common needs at the start of their programme to further develop knowledge and skills to meet the standards of the TU Delft and its research- and design-oriented MSc programmes. The MUDE is one integrated module in which topics related to data, modelling and uncertainty quantification are applied to real engineering challenges. This module comprises two interlinked parts. The Theory, Application and Coding (TAC) part focuses on teaching and applying the fundamental concepts on modelling, uncertainty and data (MUD), as well as coding skills. In the Q1 Project part students work in multidisciplinary teams on cases in the context of a smart society, where they will apply the knowledge and skills from the TAC. In the Q2 project, the students work at the interface areas where the three topics overlap, creating opportunities for more integrated applications and the ability to focus on fields of interest per programme (when needed) while satisfying the same set of learning objectives. A gradually increasing complexity and openness of inquiry will be applied.	
Study Goals	After successfully completing the MUDE a student General - is able to describe and formulate a research question (or alternatively, design requirements) given a set problem and select the appropriate methodology and tools - is able to present a fitting work plan to investigate a set of research questions or design requirements - is able to compose a technical document using appropriate academic language and citation with references - is able to work in a collaborative group environment effectively - is able to code according to basic coding standards (e.g., consistency, readability, conciseness, structure, etc.) and collaborate with their peers via distributed control software (e.g., git) - can perform spatial / temporal / multivariate analysis of data to extract knowledge via physics-based modelling tools, data-driven approaches, and uncertainty quantification methods - can present and communicate with peers in Civil Engineering and Geosciences, results of analyses using specific modelling, uncertainty and data approaches with appropriate metrics and visualization techniques Modeling - can design a modelling framework (from problem conceptualization to governing equation setup) for a physical/engineering process - can translate the modelling framework into discretized equations and computer code - can mathematically formulate and solve an optimization problem and discuss its properties - can assess optimization and simulation models performance using a set of indicators Uncertainty - can derive relevant models and complete probability and statistic calculations for faculty-wide applications - can construct relevant models for probabilistic dependence, for example using bivariate distributions or discrete Bayesian	

	Networks - use deterministic models with probabilistic inputs to evaluate engineering questions to quantify model output uncertainty (in time and/or space) - use risk and reliability analyses concepts to describe systems, their characteristics and behaviour in time and space to support decision making under uncertainty Data - can identify parameters of interest, describe the functioning of common field/remote sensors used for measuring them, and explain resolution implications in space and time - is able to apply parameter estimation using observations and perform a quality assessment - can illustrate the procedure of gathering data from different sources, pre-processing it for analysis purposes, and sharing it according to FAIR principles
Education Method	Each week starts with a plenary session to engage students by introducing the topic and showing the relevance. There will be three blocks per week to work on Theory, Application and Coding. In each block, students will have to prepare themselves by watching videos / reading some documents / doing exercises. Support will be available. Next follows an interactive class session, where teachers and teaching assistants are available to discuss and review the materials, and to work on assignments. In addition, students spend one block on the project assignment.
Assessment	Formative assessment based on Jupyter Notebooks, in-class exercises and project deliverables. Summative assessment: - Exam at end of Q1 (resit in Q2) (25%) - Exam at end of Q2 (resit in Q3) (25%) - Project portfolio (50%) You will pass the module if you obtained at least a 5.8 for the three individual components. There will be a resit opportunity for the exams. If the grade for your project portfolio is below 5.8, you will get the opportunity to submit a revised version for which you will get a 6.0 if judged sufficient.
Expected prior Knowledge	University units, or equivalent, in Calculus, Linear Algebra, Probability & Statistics, Numerical Analysis, Python programming
Academic Skills	Analysis, coding, communication, visualization, project management, team skills
Literature & Study Materials	See Brightspace
Judgement	Formative assessment based on Jupyter Notebooks, in-class exercises and project deliverables. Summative assessment: - Exam at end of Q1 (resit in Q2) (25%) - Exam at end of Q2 (resit in Q3) (25%) - Project portfolio (50%) You will pass the module if you obtained at least a 5.8 for the three individual components. There will be a resit opportunity for the exams. If the grade for your project portfolio is below 5.8, you will get the opportunity to submit a revised version for which you will get a 6.0 if judged sufficient.
Permitted Materials during Exam	Pocket calculator, formula sheet
Collegerama	No

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE Programme: Principles of Environmental Engineering

ENVM1000	Principles of Environmental Engineering	9
Responsible Instructor	Dr.ir. M.W. Ertsen	
Instructor	Dr. E. Mostert	
Instructor	Dr. J. Gebert	
Instructor	Dr. B.M. van Breukelen	
Instructor	Prof.dr. A. van de Wiel	
Instructor	A.R. Gammon	
Contact Hours / Week x/x/x/x	14/x/x/x 8(seminar)/4 (practicum)/2 (project)/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	Today's societies are facing multiple environmental changes of natural and anthropogenic origin that can both impair human and environmental health. The main priority of environmental engineers is to safeguard environmental quality, human livelihood, health, well-being, and safety. To design effective engineering and management solutions, environmental engineers need to understand both physico(bio)chemical processes in soil, water, and air as well as the societal processes that affect environmental legislation and courses of action. These two aspects are covered within the themes Environmental Quality, Cycles and Fluxes and Environment and Society. Environmental Quality, Cycles, and Fluxes Link to learning objectives of the module: ENV-PC: LO1, LO2, LO3, LO4, LO5 and LO8ab. Regarding physico(bio)chemistry, this module provides the student with quantitative insights into the interlinkages between the soil, water, and air compartments, specifically the processes distributing water (the hydrological cycle) and elements (for example C, N, P, S cycles). The module will also discuss how human activity influences these cycles and impairs soil, water, and air quality. It is explained how various environmental quality standards protect environmental and human health. Environment and Society Link to learning objectives of the module: ENV-PC: LO1, LO4, LO5, LO6, LO7 and LO8ab. Several different Environmental Engineering problems will be selected to show the variety of environmental engineering problems and how they could be analysed and solved. Problem ownership and responsibility will be discussed together with distribution of costs and benefits and ethical dilemmas. Students learn about the role of law and politics at different levels. With the gained understanding and skills, the student is able to identify integrated strategies to practically prevent or resolve environmental engineering problems. Other The students will start compiling their portfolio that will be assessed in Q6. An introduction will be given on the portfolio compilation, life-long learning skills and the ethics learning line.	
Study Goals	After completion of this module the students are able to: ENV-PC-LO1: Understand the natural cycles of water, carbon, nitrogen, phosphorus, and sulphur, and how human activities interfere and may cause environmental problems. (ILO: a) ENV-PC-LO2: Describe the key water, air, and soil quality parameters and their interrelations, explain possible causes of their variations, how these variations relate to natural processes and human interventions, and impact human health and the environment. (ILO: a) ENV-PC-LO3: Understand conceptually how water, solutes and gases are transported in the environment, how the respective fluxes can be quantified, and how contaminants affect the quality of water, soil, and air. (ILO: a) ENV-PC-LO4: Explain how to collect water, air, and soil samples; analyse those on relevant parameters; and provide a basic Interpretation of these data. (ILO: e) ENV-PC-LO5: Assess the role of stakeholders (individuals, private sector, government bodies) and the contribution of environmental standards, wider legislation, and politics in solving environmental problems. (ILO: h) ENV-PC-LO6: Identify and analyse ethical dilemmas concerning environmental problems and solution strategies. (ILO: l) ENV-PC-LO7: Design a draft strategy for solving environmental problems that integrates both understanding of the technical aspects of the problem and of the legal, political, and ethical constraints and possibilities. (ILO: a, l) EE-PC-LO8a: Perform effectively in a team of environmental engineering students that explore and analyze environmental engineering problems (ILO: k) EE-PC-LO8b: Prepare effective communication products to share the results of an environmental engineering project (ILO: i)	
Education Method	The efforts of students will strongly focus on the integration learning goals. Students teams work on real-life issues in student teams. In the teams, students apply aspects that are offered through a series of separate sessions on states/fluxes, environment&society and measurements to their project setting. This includes working on questions including: For whom is what the problem? What are related fluxes/stocks? Who is responsible for which part of the process? What data are used to act on this responsibility? How can these flows be measured? What are potential solutions? The group prepares a set of documents focusing on a feasible strategy for the given setting: a description of the origin, pathways, destinations and effects of the substance concerned, an identification of the stakeholders concerned, and selected actions that the group proposes plus a justification why to take these actions. In the last few weeks, students elaborate on their individual additions to the group work. Students start compiling their portfolio with information on ethics and lifelong learning as well as proof for fulfilling their ILOs. More information on the portfolio can be found in the Thesis preparation & Portfolio module.	
Assessment	During the sessions and project meetings, several formative formats will be applied: Quizzes on Brightspace and interactive discussions in class Discussion Forum on Brightspace	

	Short exercises with immediate feedback during class Peer feedback on draft chapters and reports on field and lab practicals Feedback from teachers on presentations and texts before handing in final versions. Practice exam and Q&A before the exam session Interviews on the student portfolio Students complete the module successfully when passing the 2 summative (individual) assessments: Written Exam fluxes/states (including measurements) = 40% Project (integrating fluxes/states and env/soc = 60% A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Expected prior Knowledge	NA
Academic Skills	Communication and responsiveness, scientific writing and presentation, time management and anticipation, application of theory to new situations academic reading, writing, referencing, information collection and evaluation, inquiry thinking, critical thinking, ethical reflection, and teamwork.
Literature & Study Materials	Short manual with links to study material prepared by the teaching teams, scientific papers, chapters/sections in open access e-books, and links to high quality information on internet.
Judgement	See Assessment
Permitted Materials during Exam	TBD
Collegerama	No

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE Track Atmospheric Environment Engineering (AEE)

ENVM1700	Fundamentals of AEE	15
Responsible Instructor	S. Basu	
Instructor	Prof.dr. A. Metrikine	
Instructor	Dr.ir. O.A.C. Hoes	
Instructor	Dr. G. Biskos	
Instructor	A. Tsouvalas	
Instructor	S. Basu	
Instructor	Dr. A. Cabboi	
Instructor	Dr.ir. B.J.H. van de Wiel	
Instructor	Dr. A. Cicirello	
Contact Hours / Week x/x/x/x	Urban climate physics: Q2: W1-7, HC(4), PT(4); W8, S(1) Atmospheric chemistry: Q2: W1-8, 4 HC (2x2) Air pollution meteorology & dispersion: Q3: W1-8, 4 HC (2x2) Physics of sound and human perception: Q3: W1-8, 4 HC (2x2)	
Education Period	2 3	
Start Education	2	
Exam Period	2 3 4	
Course Language	English	
Course Contents	This module is aimed to teach students the fundamentals of physics and chemistry in the urban atmospheric environment. They will learn to appreciate the inherent complexity of the urban environment, where due to complex building geometry the atmosphere is highly non-uniform and transient in space and time. By performing order-of-magnitude (scaling) analysis of various physical and chemical processes, the students will learn to simplify complexity into a limited number of key-processes and to describe the evolution of relevant physical/chemical quantities using the (simplified) governing equations. On an introductory level, the student will learn how those quantities can be probed by simple measurement principles. They will also learn basic tools of numerical modelling/conceptualization to analyze the impact of novel engineering designs on urban atmospheric environments. While both measurement and modelling techniques are introduced on a basic level here as to understand the key processes, an in-depth treatise of these techniques are introduced in the next module. Urban Climate Physics Link to learning objectives of the module: SS: LO1, LO2, LO3, LO4, LO5, LO6 and LO8. The module Fundamentals requires a basic introduction to urban microclimate. Here we introduce a range of physical concepts that are utilized in the remainder of the track modules. Apart from theory, some simple experiments are performed as an introduction to more rigorous explanation of experimental principles in later modules. The central theme of Urban Climate Physics is the scientific analysis of the urban climate. We aim at understanding the dominant physical processes that occur in urban environments as well as to understand the interactions between a city and the overlying atmosphere and how it impacts our living environment. Special emphasis is given to quantification of thermodynamic processes, such as radiative, convective and conductive heat transport and to airflow in complex geometries. The theoretical part connects with short practical's which illustrates the various physical principles through simple hands-on experiments. These practical's facilitate students to develop scientific intuition for the processes at hand and prepare them for the rest of the module (Modelling of Urban Flows, Urban & Indoor Air Quality, and Urban Extreme Weather). The following topics will be addressed: Introduction and basic concepts; Research Methodologies; Transport phenomena and urban air flows (part I); Transport phenomena and urban air flows (part II); Radiative processes; The urban energy balance (including moisture effects); Human thermal comfort in cities. Atmospheric chemistry Link to learning objectives of the module: SS: LO1, LO2, LO3, LO4, LO5, LO6 and LO8. In Atmospheric chemistry a range of fundamental concepts for understanding the key chemical processes that take place in the atmospheric environment will be introduced. These concepts will allow the students to identify, analyze, and explain certain atmospheric phenomena including physicochemical reactions, that will prove to be highly useful for the rest of the module. Apart from theory, simple experiments will be included during which the students will be able to get hands-on experience. The following topics will be addressed: Background, composition, and structure of the atmosphere Concepts of atmospheric chemistry (thermodynamics, kinetics, lifetime, photochemistry, etc); From stratospheric to tropospheric chemistry; Chemistry in urban and rural environments; Basic properties of atmospheric aerosol particles; Processes involving atmospheric aerosols*; Impacts of air pollution on human health, the environment and climate. Course Contents Continuation Air Pollution Meteorology & Dispersion Link to learning objectives of the module: SS: LO1, LO2, LO3, LO5 and LO6. In Air Pollution Meteorology & Dispersion the fundamental concepts related to turbulent diffusion and dispersion are introduced. The more advanced Urban & Indoor Air Quality, Integrated AEE Project will leverage on these concepts. Air Pollution Meteorology & Dispersion provides an in-depth understanding of how various atmospheric processes impact air pollutant transport, especially in urban environments. First, a number of fundamental concepts related to diffusion and dispersion (e.g., turbulence mixing, plume rise, deposition) are introduced. Then, different types of pollutant sources (e.g., point, line, area)	

and associated mathematical equations for diffusion are derived from the first principles. More advanced approaches, based on statistical and similarity theories, are discussed next. As complementary to the theoretical discussions, a state-of-the-art dispersion modelling software is utilized for further understanding of real-life pollutant transport problems. Lastly, accidental release of hazardous materials in mega-cities and related emergency response management will be briefly touched upon.

The following topics will be addressed:
 Introduction; Gaussian diffusion models;
 Plume rise, settling, and deposition;
 Statistical theories of puff and plume diffusion;
 Similarity theories of turbulence and diffusion;
 Modelling dispersion at city scale;
 Impacts of highways and buildings on urban air quality;
 Accidental releases and emergency response management.

Physics of Sound and Human Perception
 Link to learning objectives of the module:
 SS: LO2, LO3, LO5, LO6 and LO8.

In Physics of Sound and Human Perception the fundamental concepts related to physics of sound and its effect on people are introduced. The more advanced Noise Modelling & Mitigation, Integrated AEE Project will build on these concepts.

Firstly, the physics of sound are introduced with a combination of practical experimentation and mathematical analysis. Basic examples such as open and closed tubes, and moving sources are studied to provide a solid foundation of acoustical understanding in preparation for the rest of the module. Then, the way we hear and respond to sound, the way we quantify sounds, and the relationship between those measures and human responses to the sounds are introduced.

More specifically the following topics will be covered in a number of teaching sessions combining theory and (in-class) experiments related to sound and its perception:

- Derivation of the one-dimensional (1D) wave equation for a fluid-filled pipe based on first principles and explanation of the assumptions necessary to do so;
- Evaluation and analysis of free vibrations of a fluid-filled pipe: the physical phenomena of resonance and standing waves in a fluid-filled pipe;
- Derivation of the three-dimensional wave equation based on first principles and explanation the assumptions necessary to do so;
- Explanation of the physics of the wave equation in application to acoustic problems;
- Explanation of the meaning of common terms in wave mechanics, such as frequency, wavelength, wavenumber, wave speed, diffraction, reflection, dispersion and give examples of how they apply to sound;
- Acoustic sources;
- Speed of sound in a fluid medium and dependence on its physical properties;
- Demonstration (by means of a computer program where appropriate) of standing waves, interference fields, and other wave phenomena;
- Parts of the outer, middle and inner ear and their role in hearing;
- Hearing of an individual;
- Analysis of sound measurements;
- Conversion of narrowband data into third and octave band levels either with linear or A weighting.

Study Goals

After completion of this module the students are able to:

SS-LO1: Describe the overall physical and chemical structure and characteristics of the urban atmosphere and be able to express its difference with the rural surroundings. (ILO: a)

SS-LO2: Simplify the complexity of urban environments and to define the key physical and chemical concepts that describe the urban environment. (ILO: a, f)

SS-LO3: Quantify those physical and chemical concepts through an understanding of the underlying governing equations. (ILO: a, f)

SS-LO4: Perform simple hands-on experiments to measure various physical/chemical aspects of urban environments. (ILO: e)

SS-LO5: Analyze observational data of urban environments and be able to quantify its uncertainty. (ILO: e)

SS-LO6: Perform numerical modelling to quantify the effects of novel engineering designs (e.g., green roofs, parks, sound barrier) on urban environments. (ILO: d)

SS-LO7: Reflect on the societal impact of interventions and be able to discuss ethical aspects of those within the context of atmospheric environment engineering. (ILO: i)

SS-LO8. Describe how atmospheric environment pollution (e.g. related to aerosols, noise pollution) affects human health. (ILO: b)

Education Method

Teaching and learning that encourages students to take responsibility for their learning process:

Urban Climate Physics

Lectures. Prior to lectures corresponding book chapters/handouts should be studied. To engage the students, short mini exercises are given during the lectures. In addition, table-top experimental demonstrations are given during the lectures which require active student participation.

Hands-on experiments on campus. Students perform small practical's to learn basic principles of airflow & mass-conservations, radiative transport, atmospheric moisture, etc.

Article discussion in class. Students are encouraged to provide related articles from the news or from popular scientific reading, which then are discussed in class.

Instructions and theoretical exercises. On campus instructions are given, as to practice exercises. Feedback is provided by the teachers and teaching assistant during the instructions. Students are encouraged to work in small groups to stimulate discussions. To facilitate self-study, apart from direct feedback, written answers are also provided.

Prior to the exam a question/answer session is organized to better prepare for the exam.

Atmospheric chemistry

Form(s) of instruction, individual and/or group work: lectures and group work where they will carry out specific exercises and in which all the students should participate. Groups will contain 2-3 members.

Required preparation and required effort: The students are expected (i) to read background material in the form of key papers related to the topics covered each week and present the material in groups; (ii) carry out basic calculations related to the context of the lectures, and work through these with the class each week.

On campus and online activities: Online lectures and group interaction will apply only if restrictions are in place.

Air Pollution Meteorology & Dispersion

In-class lectures will be the primary mode of instruction. In one of the lectures, the students will be given a hands-on introduction to a dispersion modeling software. In addition, throughout the course, the students will be required to read relevant journal articles and critique certain aspects of the papers. They will form small discussion groups during lecture hours and will brainstorm to come up with a different scientific approach/solution.

Assessment	Physics of Sound and Human Perception In-class lectures will be the primary mode of instruction. Students will be guided through the theory combined with in-class experiments for demonstration purposes of the basic concepts of acoustics. Urban Climate Physics Exercises to actively understand and apply lecture material - are made under supervision of the instructors (this may include usage of a simple, interactive computer-program to visualize the result; no specific computer skills are required). All answers to exercises are provided after completion. Feedback to the experimental work by the students is given both on performance and on reporting. Atmospheric chemistry Feedback from presentation and group calculation work; formative multiple-choice questionnaire and short exercises provide understanding of distinct lectures. Air Pollution Meteorology & Dispersion Every week, there will be a muddiest point activity in class. The students will be asked to create a list of the unclear or confusing concepts from the previous lecture and ongoing assignment. Such written feedback from the students will be fully anonymous and the lecturer will address them during the following class. The lecturer might re-teach some of these difficult concepts with different illustrations in order to improve student learning. There will be three individual assignments which will empower the students with the necessary understanding and skill sets to execute a class project. The assignments will be comprised of theoretical problem solving, data analysis (e.g., analysis of air quality index), and small-scale model development (e.g., Gaussian plume model, mixed layer model). The students will receive formative feedback from the instructor. In addition, there will be a weekly in-class quiz (non-graded) activity. This activity will be interactive in nature and will make use of an online polling website or a cell phone app. Physics of Sound and Human Perception During the teaching period, a number of small assignments/exercises will be distributed as practicing material so that students get familiar with the theory of sound and its perception by humans. Formative feedback will be provided to students who decide to deliver assignments by both the instructor and the peers. Students complete the module successfully when passing the following 4 summative assessments: Urban climate physics: final written exam with closed and open-ended questions (33%) Atmospheric chemistry: written exam (27%) Air pollution, Meteorology and dispersion: individual assignment report (20%) Physics of Sound and Human Perception: written exam (20%) A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Expected prior Knowledge	The pre-requisites are inline with the admission criteria. The students should have prior knowledge of mathematics (calculus, linear algebra, differential equations, probability and statistics, numerical methods), fluid mechanics, and python programming language.
Connection to other modules	This foundational module will enable the students to learn more advanced materials in subsequent modules.
Academic Skills	Communication and responsiveness, scientific writing and presentation, time management and anticipation, application of theory to new situations using the relevant equations, academic reading, writing, referencing, information collection and evaluation, inquiry & critical thinking and teamwork.
Literature & Study Materials	Urban Climate Physics The book: Urban Climates by Oke, Mills, Christen and Voegt. Cambridge University Press, Un. Printing House, Cambridge CB2 8BS, U.K. Additional background material and/or papers will be provided via BrightSpace. Experimental material is provided in the practical's. Atmospheric chemistry One main textbook: Atmospheric Chemistry and Physics: From Air Pollution to Climate Change, J. H. Seinfeld and S. N. Pandis* Background material, papers and chapters from relevant literature will be provided via BrightSpace. *Ask Prof. Spyros Pandis to come as a guest lecturer to introduce PMCAMx; one of the main algorithms for predicting regional air pollution Air Pollution Meteorology & Dispersion Textbook: de Visscher, A. (2014), Air Dispersion Modeling: Foundations and Applications, John Wiley & Sons, Inc., 634 pp. Reference book: Arya, S. P. (1999), Air Pollution Meteorology and Dispersion, Oxford University Press, 310 pp. On a regular basis, the students will be provided with various journal articles related to the course topics. It is anticipated that they will peruse the articles before class and participate in class discussion. Physics of Sound and Human Perception Background material, papers and chapters from relevant literature will be provided via Brightspace. Material will be borrowed by the following books: https://www.taylorfrancis.com/books/fundamentals-sound-vibration-frank-fahy-david-thompson/e/10.1201/b18348 https://www.springer.com/gp/book/9783540240525 https://www.springer.com/gp/book/9783030447861 https://www.cambridge.org/core/books/fundamentals-of-noise-and-vibration-analysis-for-engineers/8768D986559E0B54ECDDF0A167C9CDF1 https://www.springer.com/gp/book/9783662053911
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	During open book exams, the students are permitted to bring any written materials.
Collegerama	Yes

ENVM1800	Atmospheric Measurements and Modelling	9
Responsible Instructor	Dr. S.R. de Roode	
Instructor	C.M.H. Unal	
Instructor	Dr.ir. J.A.E. ten Veldhuis	
Instructor	Dr. G. Biskos	
Instructor	Dr. S.R. de Roode	
Instructor	A. Tsouvalas	
Instructor	S. Basu	
Instructor	Dr. M.A. Schleiss	
Instructor	C. Garcia Sanchez	
Instructor	Prof.dr.ir. R. Uijlenhoet	
Contact Hours / Week x/x/x/x	6 (HC)/4 (PC)/2 (PT)/10 (PJ) /x/x/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This module provides in-depth educational training on the principles of measurement and modelling techniques with a focus on urban environments. Through various activities, the students will gain significant hands-on experience in conducting field experiments and performing numerical simulations. In collaboration with ethics experts from the TPM faculty, the instructors of the AEE track will organize a half-day long workshop on ethics. The theme of the workshop will be on ethical issues related to Open Science. The students will actively participate and prepare a reflection document as part of a portfolio. During Q6, this portfolio will be evaluated by ethics experts. Atmospheric Measurement Techniques for Urban Environments This module theme covers the fundamentals of observational techniques for atmospheric processes with an emphasis on urban environments and high spatial/temporal resolutions. The following topics will be addressed: Importance of collecting data Difficulties encountered in urban environments General overview of different types of measurement techniques including Doppler and polarimetry Basic concepts: wavelength, scattering, absorption, propagation Basic definitions and concepts: scale, coverage, temporal/spatial resolution, accuracy, precision etc. Rain gauges, Radar, Turbulent flux measurements of momentum, heat and moisture and chemical species, Lidar, Sodar, Radiometers, Accuracy, precision, calibration Noise measurements in the field, analysis of measured data and filtering Quality control, noise, bias, uncertainties and errors. Missing values Sensor placement, good practices etc. Citizen weather stations Microwave links, smartphones. User engagement and awareness. Modelling of Urban Flows Content module theme: The module theme focuses on the fundamentals of Computational Fluid Dynamics (CFD) tailored for urban environments. The course will introduce the governing equations first and elaborate on the mathematical methods applied to model the flow around buildings. Real world examples will be used to show direct application of CFD during computer labs, and students will have hands-on practice to learn and use CFD in urban areas. The following topics will be addressed: Governing equations Reynolds decomposition and averaging Discretization of governing equations (finite-difference, finite-volume, finite-element approaches) Time-marching methods Stability, accuracy, convergence Boundary conditions for urban environments Mesh and complex geometries generation Turbulence modelling (RANS/LES/DNS) Introduction to uncertainty quantification	
Study Goals	Measurements: LO2, LO3, LO4, LO6 and LO7. Modeling: LO1, LO2, LO3, LO4, LO5, LO6 and LO7.	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process: Atmospheric Measurement Techniques for Urban Environments Lectures + in-class activities + group project List of topics for projects: Topic 1: noise measurements on campus. Topic 2: measurement of temperature, wind and rainfall on campus using Davis and Netatmo weather stations Topic 3: radiometer and pyranometer data analysis Topic 4: heat stress and urban heat island analysis Topic 5: rainfall estimations using radar Topic 6: sodar-based wind data analysis Topic 7: indoor/outdoor air quality measurements Topic 8: citizen weather data analysis and quality assessment Topic 9: rainfall estimations using microwave links Topic 10: wind estimations using radar Modelling of Urban Flows This theme includes lectures and computer labs in which students work in groups as well as individually.	

	During lectures active learning activities are included to keep all the students on the same page, and enhance participation. Some examples include: - flipped classroom, where materials would be provided for the students to read before the class (for example when covering CFD domain size design we will use published literature); - in-class, online quizzes after covering theoretical concepts; - students will be asked to associate physical processes with various terms of the fluid dynamics equations; 5 minutes notes at the end of the theoretical lectures to collect the most difficult concepts of the session. During computer labs, students will perform CFD simulations. Several topics will be provided with solutions to train for the final group project. During computer labs, they will work on their own simulations and the teacher will be there for troubleshooting. Some potential modelling topics are: - wind flows around high-rise buildings; - dispersion of passive scalars; - wind flows around trees; - wind flows over heterogeneous roughness areas. On campus and online activities Classes will take place online if on campus is not possible
Assessment	Atmospheric Measurement Techniques for Urban Environments All in-class activities (including the practical's and homework) will be formative. Students will receive peer-feedback during the in-class activities and formative written feedback on the homework from the TA's. Modelling of Urban Flows Formative feedback during in-class activities, with interaction between peers. Formative assignments will be proposed to the students during laboratories, with feedback provided within 2 weeks. During the first week, their peers will have the opportunity to provide feedback. During the second week, the teacher will provide feedback. Students complete the module successfully when passing the 3 summative assessments: - Atmospheric Measurement Techniques for Urban Environments: written exam (30%), group project (30%) - Modelling of Urban Flows: group assignment (report (35%) and presentation (5%)). Groups will be composed by 2-3 students (depending on the group size and computational resources). We will use buddycheck to evaluate the group participants contribution.
Connection to other modules	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment. The AEE track will start with only one module A.
Academic Skills	Problem-based learning and solving, making inferences of physical situations by analyzing observational and modelled data, academic reading, scientific writing, referencing, inquiry thinking, critical thinking and teamwork.
Literature & Study Materials	Atmospheric Measurement Techniques for Urban Environments - Introduction to Meteorological Instrumentation and Measurement, by Thomas P. deFelice. - Peer-reviewed articles and review papers will be provided through Brightspace. Modelling of Urban Flows Slides will be provided for the students before every lecture. Additional materials as papers or content not included in the reference book will be provided for the students. Most lectures will be based in the reference book mentioned hereafter. Reference book: - Computational Fluid Dynamics: A practical approach, Jiyuan Tu, Guan-Heng Yeoh, Chaoqun Liu. - Additional cluster resources will be necessary to run CFD simulations (Delft cluster/INSY)
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations

ENVM1900	Grand Challenges in AEE	15
Responsible Instructor	Dr.ir. B.J.H. van de Wiel	
Instructor	C.M.H. Unal	
Instructor	Dr.ir. O.A.C. Hoes	
Instructor	Dr.ir. J.A.E. ten Veldhuis	
Instructor	Dr. G. Biskos	
Instructor	Dr. S.R. de Roode	
Instructor	Dr. A.A. Nunez Vicencio	
Instructor	A. Tsouvalas	
Instructor	M.M.E. van Esch	
Instructor	Dr. K. Anupam	
Instructor	Dr. Z. Yang	
Instructor	S. Basu	
Instructor	Dr. M.A. Schleiss	
Instructor	H.O. Sertlek	
Instructor	C. Garcia Sanchez	
Instructor	W.T. Ball	
Instructor	Dr. A. Cicirello	
Contact Hours / Week x/x/x/x	14 (HC)/ 4 (PT)/ x/x/x/x	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The previous modules focused on understanding, conceptualization and technical aspects of Atmospheric Environment Engineering (AEE). This module takes a step beyond. Students first learn to identify and analyze grand challenges within AEE. Next, students learn how to create and evaluate engineering solutions for such challenges of contemporary and future societies. Rather than working on an isolated technical aspect, they learn how to integrate knowledge of different disciplines and learn to integrate different tools to tackle the problem at hand. To monitor and manage a research project and to work in an interdisciplinary team. To communicate their scientific and engineering solutions and to reflect on the ethical implications of their proposed solutions to the grand challenges. In collaboration with ethics experts from the TPM faculty, we will organize a half-day long workshop on ethics. The theme of the workshop will be on ethical issues related to Public Health. The students will actively participate and prepare a reflection document as part of a portfolio. During Q6, this portfolio will be evaluated by ethics experts. Urban and Indoor Air Quality LO1, LO3, LO4, LO7 and LO9. This course provides an in-depth understanding of how various sources of pollution can affect air quality in the urban outdoor and indoor environment. The course will start by covering the different types of air pollutants (gaseous pollutants, particulate matter, and bioaerosols), their sources, as well as the processes they may undergo to lead to secondary pollutants. Links with the associated health effects will be made. We will introduce simple observational and computational techniques for assessing and predicting the evolutions of air quality in the outdoor environment. We will draw upon the knowledge of module A in order to demonstrate how the concentration of different air pollutants can be monitored in the outdoor and indoor environment. Simple computational models will be introduced for the case of the indoor environment, showing the students how different ventilation approaches and other mitigation measures (humidification/purification) can improve air quality in buildings and/or vehicles. Topics of special interest on how different air pollutants, including particles containing viruses, can disperse/transport in the indoor environment will also be covered (guest lecture). Noise Modelling and Mitigation LO1, LO2, LO3, LO4, LO6 and LO9. We first introduce computational methods available to solve practical problems in acoustics and compared them with benchmark analytical solutions. Potential and limitations of analytical and numerical approaches are discussed. The module units introduces available methods but also stimulate to think beyond for those seeking to apply or extend them further. The second part of the module theme deals with noise control engineering, focusing primarily on noise mitigation and noise trapping, to reduce noise in practical situations. The noise control engineer needs to know how to set targets, how to characterize and quantify noise sources, and how to reduce noise either at source or in the transmission path. Suitable formulae are provided and explained for this. The main assessment consists of a design calculation/field study which makes use of the above to solve a practical problem	
Course Contents Continuation	Extreme Urban Weather LO1, LO2, LO3, LO4, LO6 and LO9. The first part focuses on rainfall extremes. Students learn how environmental factors such as temperature and relative humidity influence the spatial/temporal distribution of rainfall. They learn how to use radar and numerical weather prediction models to predict rainfall over urban scales. This enables them to identify dangerous rainfall patterns, assess risks and discuss the effectiveness of mitigation strategies. The second half of the module theme focuses on modelling the urban microclimate on the scale level of the street. This enables students to indicate problematic areas in relation to heat stress, as well as propose and assess the effectiveness of mitigation strategies. The following topics are addressed: Scaling of precipitation with temperature. Extreme precipitation in numerical weather prediction models; Nowcasting using radar Droughts & dry spell analysis; Influence of urban environment on heat stress and mitigation measures; Modelling urban heat using ENVI-met; Presentations from experts in the field of climate adaptation. Two assignments will run in parallel: a re-analysis of a heavy rain and a heat stress event (including analysis of potential mitigation)	

Integrated AEE Project
LO1, LO2, LO3, LO4, LO5, LO6, LO7, LO8 and LO9.

This module theme aims to integrate knowledge acquired during previous modules. Here we go beyond theoretical understanding and application of techniques by analyzing a complex research question as a whole. The student will break down the complex problem in sub-parts and will identify which methods/techniques are most appropriate to answer the research question. New ideas will be developed and new insights will be generated by combining or improving knowledge from different sources, which may include novel data analysis, modeling or theoretical analysis. The results are placed within a societal context and environmental and/or ethical issues are taken into consideration. Students will therefore condense the scientific outcome into a summarized form displaced in poster format. The posters will be presented to a broad audience and will strive to address the critical questions posed by the audience. Finally, the student will reflect on the full cycle of his/her research within his/her peer group as to learn how to improve his/her approach to this type of assignments in the future (master thesis).

Examples of projects:

- Understanding and prediction of intra-diurnal pollution level variations in a large city. As both human activity (rush hours) and atmospheric activity (turbulent mixing) strongly varies within a circadian cycle, this is also true for pollution levels exposed to the inhabitants.

- Harvesting the Urban Wind Island: a potential source of wind energy? The wind over city regions with low buildings may exceed wind over their rural counterpart that are less rough in general. This counterintuitive aspect results from the fact that the atmospheric boundary layer is generally deeper over cities.

- Greening the city: the role of vegetation in climate adaptation. Trees, shrubs and grass bring many benefits to cities including cooling by transpiration, filtering aerosols, absorbing noise and enhancing the general well-being of the urban population. Yet urban vegetation lives under stressful conditions. The project can take different angles. E. g by observing and modeling of temperature, radiation, noise, air quality modification. Or by looking into the design of urban greening by finding sweet spots in urban microclimate.

Study Goals

After completion of this module the students are able to:

B-LO1: Identify grand challenges in atmospheric environmental engineering related to sustainability of the built environment and to human health. (ILO: b)

B-LO2: To translate these problems into scientific research and/or engineering questions. (ILO: b)

B-LO3: Select the appropriate scientific tools (theoretical/ observational/ numerical) to address those questions and become cognizant of the limitations of the chosen tools. (ILO: a)

B-LO4: Apply various scientific tools in a rigorous and efficient manner. (ILO: e, f)

B-LO5: Plan, monitor and manage a project and to collaborate in an interdisciplinary team with people from diverse backgrounds. (ILO: j, k)

B-LO6: Develop novel ideas for interventions; existing knowledge is critically assessed, and alternatives are sought for by integrating multi-disciplinary approaches. (ILO: c)

B-LO7: Reflect on the societal impact of interventions and be able to discuss ethical aspects of those within the context of atmospheric environment engineering. (ILO: l)

B-LO8: Effectively communicate research findings to other researchers, engineers, as well as to the general public. (ILO: i)

B-LO9: Demonstrate understanding of the following UN sustainable development goals: (a) good health and well-being; (b) affordable and clean energy; (c) sustainable cities and communities; and (d) climate action. (ILO: h)

Education Method

Teaching and learning that encourages students to take responsibility for their learning process:

Urban and Indoor Air Quality

In-class lectures will primarily cover the necessary background knowledge so that the students will be capable to collect necessary data and use appropriate models to assess air quality in the indoor and outdoor environment. In a small number of the lectures the students will be given hands-on instruction on how to use specific sensors and models. They will get the opportunity to employ these tools at a later stage in order to carry out an assignment. In addition, throughout the course, the students will be required to read relevant journal articles and critique certain aspects of the papers in groups during the lecture hours.

Noise Modelling and Mitigation

Class lectures (on-campus) will be the primary mode of instruction to convey the theoretical concepts of the module. Students will be guided through the theory combined with in-class experiments.

Note: This theme will make use of collected noise measurement data from the theme Atmospheric Measurement Techniques for Urban Environments in which students will work on a field assignment which includes the measurement and analysis of recorded sound signals.

Extreme Urban Weather

In class lectures + practicals;
Assignments running in parallel with lectures.

Integrated AEE Project

Lectures: 2x 2 hours of lecture is given. The first lecture is to introduce and to give an overview on potential research topics, an outline of methodologies and a presentation of learning objectives. The second lecture is an introduction on available methods/guidelines for scientific poster presentation.

Training exercise: during an afternoon several examples of characteristic posters from conferences are discussed in class. Groups of students discuss e.g., highlights & pitfalls and propose a ranking of those examples.

Group work. The project work is done in groups of 2-5 students. This strategy is to stimulate interaction among persons with different backgrounds and different characters and competences. Each group is supervised by an instructor who coaches the research process.

Mini symposium. The results are displayed in a mini-symposium (potentially with peer-students and faculty members). If desired by the students, this symposium could be done in a quasi-formal setting of a mock conference (including drinks and formal clothing).

Assessment

Urban and Indoor Air Quality

Specific questions (which will require short calculations) related to the covered materials will be provided to the students in every lecture. The answers to these questions will be discussed either in following lectures or during specific sessions that can be arranged depending on demand. In addition, feedback will be provided in the framework of the assignments, both during the explanation/training but also during the presentation of the results.

Noise Modelling and Mitigation

Formative assessment of the work-in-progress will be provided to each group in several occasions in the form of oral feedback by both the teacher and the peers.

	Extreme Urban Weather All in-class activities (including the practical's and homework) will be formative. Students will receive peer-feedback during the practical's and formative written feedback on the homework from the TA's. Integrated AEE Project Once a week the project group meets with their respective instructor/coach. Students are expected to be active: a brief update on progress is given by preparing and presenting a very short PowerPoint (few slides only). As all students should be engaged, this preparation is done on a rotational basis, with one student responsible for a specific week. This alternating strategy stimulates continuous engagement by all the students and also allows individual feedback at an early stage. During the poster presentation, direct feedback is given by the peer students (intervision). Together with the teacher the students reflect on what could have been improved in the project and what are the recommendations for future work. Summative feedback. After the poster presentation, a project grade is given by the teacher team according to (rubric) criteria that take into account various aspects like e.g.: scientific approach, theoretical depth, organization and group work, end result, written (poster) presentation, contribution to discussion and oral presentation. Students complete the module successfully when passing the following 4 summative assessments: Urban and Indoor Air Quality: group assignment (20%) (including report 15% and presentation 5%) To avoid 'freeriding', a statement is made at the end of the report which indicates specific contributions by each student. Likewise, each student orally presents part of the presentation. Noise Modelling and Mitigation: group assignment (presentation of the case study (10%) and report on it (10%)) Both the report and the presentation are group components. Free riding will be prevented as follows: All group members will be asked to present part of their group assignment. After the presentation of the group assignment, the assessor will ask each individual member of the group specific questions regarding the report and the presentation. Questions can span the whole assignment regardless of which part is presented by the individual group member. This way individual contributions to the group work will be assessed and differentiation of the grades (between group members) will be possible based on the answers provided. Extreme Urban Weather: written exam with closed and open-ended questions (20%) Integrated AEE project: group project (40%) As the emphasis of the project is on integration, connecting and collaboration, assessment will take place on group level. A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Academic Skills	Communication and responsiveness, scientific writing and presentation, time management and anticipation, application of theory to new situations using the relevant equations, presenting, academic reading, writing, referencing, information collection and evaluation, inquiry thinking, critical thinking and teamwork.
Literature & Study Materials	Urban and Indoor Air Quality Written background material, consisting of book chapters, scientific papers and in-house developed readers will be provided to the student in advance via Brightspace. Experimental material is directly provided in the practical's. Noise Modelling and Mitigation Background material, papers and chapters from relevant literature will be provided via Brightspace. Material will be borrowed by the following books: https://www.taylorfrancis.com/books/fundamentals-sound-vibration-frank-fahy-david-thompson/e/10.1201/b18348 https://www.springer.com/gp/book/9783540240525 https://www.springer.com/gp/book/9783030447861 https://www.cambridge.org/core/books/fundamentals-of-noise-and-vibration-analysis-for-engineers/8768D986559E0B54ECDDF0A167C9CDF1 https://www.springer.com/gp/book/9783662053911 https://www.springer.com/gp/book/9781475760118 https://www.taylorfrancis.com/books/environmental-methods-transport-noise-reduction-mats-nilsson-j%C3%B6rgen-bengtsson-ronny-klaeboe/e/10.1201/b17606 Extreme Urban Weather Lenzholzer, S. (2015). Weather in the City-How Design Shapes the Urban Climate. Nai 010 Uitgevers/Publishers. Willems, P. (2019): Impact of Climate Change on Rainfall Extremes and Urban Drainage Systems, IWA Publishing, Vol.11, Open access ebook at https://doi.org/10.2166/9781780401263 Additional background material and/or papers will be provided via Brightspace. Experimental material is provided in the practical's. Integrated AEE Project Book (selected chapters). Eloquent Science: A Practical Guide to Becoming a Better Writer, Speaker, and Atmospheric Scientist. By Schultz. Springer. ISBN 978-1-935704-03-4. Handouts on presenting a scientific poster. Journal papers for topic at hand, as collected by the students. Rubric for conference posters, American Meteorological Society
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE Track Water Resources Engineering (WRE)

ENVM1400	Quality, Quantity and Design, Information and Advice	15
Responsible Instructor	Dr.ir. A.M.J. Coenders	
Instructor	Dr.ir. M.W. Ertsen	
Instructor	Dr.ir. J.G. Langeveld	
Instructor	Dr. E. Mostert	
Instructor	Dr. R.R.P. van Nooijen	
Instructor	Dr.ir. O.A.C. Hoes	
Instructor	Prof.dr.ir. M. Bakker	
Instructor	Dr.ir. J.A.E. ten Veldhuis	
Instructor	Dr.ir. R.J. van der Ent	
Instructor	Dr.ir. G.H.W. Schoups	
Instructor	Dr.ir. A.M.J. Coenders	
Instructor	Dr. E. Abraham	
Instructor	M. Lauren	
Instructor	Dr. B.M. van Breukelen	
Instructor	Prof.dr.ir. R. Uijlenhoet	
Contact Hours / Week x/x/x/x	x/8/variable/x (8 contact hours pw in Q2. Q3 varies)	
Education Period	2 3	
Start Education	2	
Exam Period	2 3 4	
Course Language	English	
Course Contents	Water resources engineers develop engineering interventions to protect society from droughts, flooding and water pollution and minimize the scarcity of potable water. The design of these interventions requires fundamental understanding of processes that influence water availability and water quality. The Water Resources Engineering track starts in Q2 with the first part Quality, Quantity and Design of this stepping stone. In Q3 part 2 Information and Advice will be scheduled. Steppingstone part 1: Quality, quantity and design In this part of the module students will study water quality and biochemical processes, the natural hydrological cycle, and the man-made urban water cycle (drinking water, wastewater etc.) through lectures and design exercises of a range of water resources engineering systems (irrigation, reservoirs, polders, drinking water supply etc.). This module contains three different subthemes: Environmental Water Quality & Biogeochemical Processes, Environmental Water Quantity & Hydrological Processes, and Water resources engineering system design with distinct teaching approaches. Environmental Water Quality & Biogeochemical Processes Link to learning objectives of the module: SS: LO1 and LO4. This subtheme addresses environmental water quality and biogeochemistry aspects relevant to water resources management. Knowledge about the water quality and reaction processes and matter cycles taught in the ENV programme module will be used. Through cases students learn to describe relevant water quality and biogeochemical processes and make first order quantitative descriptions using hand calculations and open access software. Specific topics addressed include: Dissolution and precipitation of minerals; Thermodynamics and kinetics; pH/alkalinity Water-Air interaction and effects Sorption reactions: adsorption, ion-exchange Sorption reactions: surface complexation, pathogen filtration Redox reactions and microbial growth; kinetics Mass transport: Advection, dispersion, reaction, mixing Chemical modelling of water quality Interpreting and modelling of water quality data from environmental systems Environmental Water Quantity & Hydrological Processes Link to learning objectives of the module: SS: LO2 and LO4. This subtheme addresses environmental water quantity with a specific focus on hydrological and hydraulic processes, both in natural systems and man-made systems. It builds upon the knowledge on the hydrological cycle and water transport processes introduced in the ENV programme module. Through cases students learn to describe and quantify fluxes and stores in both the natural and the man-made water cycles. Topics that will be discussed are precipitation, evaporation, groundwater, drinking water, wastewater. Specific topics addressed include: Hydrological cycle in natural and man-made systems: precipitation, evaporation and vegetation, unsaturated zone, groundwater, human fluxes) Flow and discharge in rivers and open channels Flow and discharge in (pressurised) pipe systems and networks Flow at water control structures (weirs, orifices, pumps) Water resources engineering system design Link to learning objectives of the module: SS: LO3, LO4 and LO5. Process understanding is required to design water resources engineering systems. In this subtheme the functional dimensioning of several common water resources engineering systems will be discussed. Partly process understanding introduced in the abovementioned subthemes will be applied and partly new process understanding will be taught. Furthermore, first order design and dimensioning models (formulas) will also be part of the material. Specific topics addressed include: Multi-purpose reservoirs	
Course Contents Continuation		

- Irrigation systems
- Polder and drainage systems
- Urban drainage systems (piped and non-piped)
- Rainwater harvesting systems
- Piped water distribution systems
- Blue green infrastructure
- Wastewater treatment systems

Steppingstone part 2: Information and Advice

Water resources engineers advise stakeholders on solution strategies for problems such as water pollution, water scarcity, floods and limited access to safe water. They build advice on extensive problem and data analysis and consider the ethical implications of their advice.

The second part of the steppingstone module Information and Advice will take place in Q3 and consists of a deepening of the data analysis skills from the MUDE specifically for water and of an advisory group project. In this project students will advise a stakeholder on a real-life water resources engineering project based on substantial data analysis. Several projects will be offered on various scales (local to catchment) and systems (urban, rural etc.). Steppingstone 2 will be taught as an intensive course of three weeks full time, including lectures, workshops and tutorial sessions. Information and advice are divided into the lecture- and exercise-base sub-theme "Data quality and data analysis" and the project-based sub-theme "Evidence-based advice".

Data quality and data analysis
Link to learning objectives of the module:
SS: LO6, LO7 and LO8.

This subtheme focusses on understanding and predicting the spatiotemporal behaviour of water resources engineering systems based on data. Students learn about typical monitoring methods and uncertainties associated and how to use statistical techniques for data analysis and prediction. This subtheme builds upon the faculty wide MUDE.

Specific topics include:
Geographical Information Systems
Raster and polygon data
Spatiotemporal patterns in man-dominated fluxes such as (drinking) water demand
Uncertainty and propagation of errors in hydrological time series
Correlation and persistence in precipitation and other timeseries
Frequency analysis and extreme value analysis of precipitation and discharge
Methods for determining extreme (design) precipitation flood and drought conditions
Time dependent groundwater flow
(groundwater) timeseries analysis
Spatiotemporal patterns in surface and groundwater quality
Leakage detection

Evidence-based advice
Link to learning objectives of the module:
SS: LO6, LO9, LO10 and LO11.

In this advisory group project students will work in a diverse team in terms of personalities, cultures and study background and advise a stakeholder on a real-life water resources engineering project based on substantial data analysis. In addition, they will reflect on the challenges and opportunities that working in a diverse team brings and on their (ethical) professional responsibility as an engineer in such an advisory project. This part builds upon data quality and data analysis and on the experience with teamwork and ethics that students gained in the ENV programme module.

Specific attention will be paid to data availability and how to handle limited data availability. Students will practice systematic procedure to make assumptions and how to assess the effect of these assumptions on their final advice.

Specific topics include:
Problem analysis for water resources engineering students
Data availability and data scarcity
Assumption development and assessment (sensitivity analysis)
Team roles and team behaviour
SCRUM
Professional ethics on the responsible use of data
Presentation skills

Study Goals

SS-LO1: Describe and quantify chemical and biological processes in environmental water systems, including transport by advection, diffusion and dispersion, (bio) chemical reactions and conversions and dissolution and precipitation of minerals. (ILO: a)

SS-LO2: Quantify stores and (mainly stationary) fluxes in environmental water systems, including precipitation, evaporation, surface water and groundwater flows, drinking water flows, wastewater flows etc. (ILO: a)

SS-LO3: Analyse water resources engineering problems. (ILO: b)

SS-LO4: Conceptualize relevant physical, chemical and/or biological processes towards a quantitative description of environmental water systems. (ILO: f)

SS-LO5: Explain the functioning of typical engineering solutions in the field of water resources management, water treatment and urban water infrastructure. (ILO: d)

SS-LO6: Identify and analyze complex water resources problems that may threaten the environment and human wellbeing. (ILO: b, d, e)

SS-LO7: Analyse and assess the quality of observations typical in water resources engineering. (ILO: g)

SS-LO8: Analyse temporal and spatial patterns in data in water resources engineering, including extreme values to gain system understanding. (ILO: a, g)

SS-LO9: Advise on solution and/or monitoring strategies for problems in the field of water resources engineering supported with quantitative data analysis considering (inter)national goals, norms and regulations for water resources management. (ILO: c, d, h)

SS-LO10: Assess data needs and required quality based on requirements set by stakeholders involved in the problem and solution strategies. (ILO: e, h)

SS-LO11: Work effectively in a team of diverse talents, skills, characters, and cultures to solve Water Resources Engineering problems. (ILO: j, k, l)

Education Method	Teaching and learning that encourages students to take responsibility for their learning process: Part 1 The teaching method is Problem Based Learning, where students collaboratively work in small groups on several cases. Through the 7-jump method (1: clarifying concepts, 2: formulating the central case, 3: free brainstorm, 4: clustering, 5: learning objectives, 6: self-study, 7: report) students will actively construct (new) knowledge and skills to master the module content. This requires high-engagement and active participation of students. Furthermore, through the collaborative group process students can benefit from each others background. Each group has a tutor who ensures that the learning goals will be met. Study materials will be provided in different forms (text books, papers, videos, interactive lectures, (computer)practical (e.g., PHREEQC), etc.). Half-way Q2 a formative quiz will be scheduled, to give student feedback on their performance and help them to successfully meet the learning goals. Part 2 Data quality and data analysis Data quality and analysis will be taught via lectures, computer tutorials and computer exercises. Easy tutorials will be homework whereas more complex exercises will be done in class under supervision of the teaching team. Evidence-based advice Students practice the project management method SCRUM. SCRUM Teams will be formed based on diversity in personalities. Students will plan their project in three weekly sprints, organise daily stand-ups and assess weekly progress in review and retrospective meetings. Teachers facilitate this process through a series of workshops. In addition, specific guidance on problem analysis and assumption making will be offered.
Assessment	Students complete the module part successfully when passing the 3 summative assessments: - Environmental Water Quality & Biogeochemical Processes and Environmental Water Quantity & Hydrological Processes and Design (60%): Written exam - Data quality and data analysis (15%): Individual computer test - Evidence-based advice (25%): The last review and retrospective meeting will be used to assess the students using a rubric that will be developed based on the learning objectives associated with this subtheme. A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Expected prior Knowledge	Not applicable; however, having followed the Core Program Module - Environmental Engineering is highly recommended.
Academic Skills	Critical thinking; Analytical thinking; Problem-based learning; Solving; Collaboration with peers; Application of theory to new situations using relevant equations; Data analysis; Advice stakeholders; Ethical reflection/consideration.
Literature & Study Materials	Part 1: Study materials (text books, papers, videos, interactive lectures, (computer)practical (e.g., PHREEQC), etc.) will be provided via Brightspace Part 2: Data quality and data analysis For GIS students will use the free and open source program QGIS. They will be introduced via an updated version of https://ocw.tudelft.nl/course-lectures/introduction-to-gis/?course_id=9282 (accessed 2-4-2021). For the other topics dedicated lecture notes, background material, papers and chapters from relevant literature will be provided via Brightspace. Evidence-based advice Dedicated lecture notes, background material, papers and chapters from relevant literature will be provided via Brightspace, including the official Scrum guide (scrumguide.org).https://scrumguides.org/
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	None (or check Brightspace for updates)
Collegerama	Yes

ENVM1500	Water Quality and Principles of Environmental Engineering	9
Responsible Instructor	Dr.ir. R.E.F. Lindeboom	
Instructor	Prof.dr. G.J. Medema	
Instructor	Dr.ir. H.L.F.M. Spanjers	
Instructor	D.G. Weissbrodt	
Instructor	M. Laureni	
Instructor	Dr. B.M. van Breukelen	
Instructor	Dr.ir. R.E.F. Lindeboom	
Instructor	K.M. Lompe	
Contact Hours / Week x/x/x/x	x/x/4/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	To define interventions for any future water resources challenge in a systematic manner, water technology students need to be able to apply the mass balance concept and calculate the time-based performance of a system unit in terms of biological and chemical reaction rates. By starting with physicochemical processes and then expanding to biochemical processes the complexity is gradually increased while using the same methodology. Additionally, these calculations can be combined with knowledge on pollutant removal and degradation processes/factors to analyse and adapt the operational conditions of a lab experiment with the aim to optimise process performance for both efficient conversions and/or pollutant removal. The ethical aspects of working as an engineer in the field of human health will be discussed in a workshop. Principles of physicochemical and biological processes Link to learning objectives of the module: A1: LO1, LO2 and LO3. Central topic of the module sub theme is learning to apply the principles of physicochemical and biochemical processes, reactor technologies and pollutant removal/degradation processes with the aim of using this know-how for designing and evaluating a treatment unit in module B1. It provides an environmental process engineering perspective and addresses the fundamentals on: Calculating equilibria of (bio-)chemical conversion processes in natural or engineered treatment units. Assessing biological and chemical reaction kinetics in dependence to applied reactor conditions and operational process characteristics using mathematical models. Assessing removal efficiency of pollutants by physicochemical and biochemical processes, by exploring, understanding and applying underlying concepts like mass balances, (bio-)chemical thermodynamics, mass transfer phenomena, dissolution, precipitation, degradation kinetics. Biological growth models (Monod kinetics), redox processes and reaction stoichiometry. Reactor technologies. Factors that impact removal, degradation or conversion of pollutants (pathogens, antibiotic resistance, chemicals of concern) that pose a risk to human health. In addition, the ethical aspects of working as an engineer in the field of human health will be discussed in a workshop. Biological or chemical reactor performance analysis Link to learning objectives of the module: A1: LO1, LO2, LO3, LO4 and LO5. Central topic of the module sub theme is the execution of a biochemical lab experiment and mathematical modelling and performance analysis of this system. Through the lab experiment students can use know-how from module sub theme I to describe the chemical and/or biological conversion processes and biological growth rates and evaluate the removal efficiencies of contaminants (chemicals and pathogens that pose a risk to human health) in a single process unit. The following topics will be addressed: Using mathematical modelling in different software environments (i.e., phreeqC, matlab, python, aquasim) the experimental results can be described with chemical and biological conversion/growth models and evaluated for their potential to protect human health. Presenting, proposing and reporting about improved operational conditions for performance enhancement.	
Study Goals	A1-LO1: evaluate how physicochemical and biological processes in natural and engineered environments work and how they could be used to remove, degrade or recover pollutants (chemicals, pathogens, antibiotic resistance) that pose a risk to environmental and human health. (ILO: c, d) A1-LO2: Assess mass balances and (geo)chemical speciation reactions to calculate the equilibrium of (bio-)chemical conversion processes in natural or engineered treatment sub theme (batch/CSTR). (ILO: a) A1-LO3: Assess biological and chemical reaction kinetics in dependence to applied reactor conditions and operational process characteristics (Batch, fed-batch, porous media filters, CSTR, FBR/MBBR, PFR) using mathematical models. (ILO: a) A1-LO4: Execute a small lab experiment focused on a biological and/or chemical conversion technique, including a model-based prediction and verification of the results. (ILO: b, g) A1-LO5: Propose an improved approach for an environmental challenge based on laboratory results, external datasets, modelling output and scientific literature and convey one's advice through a presentation and report. (ILO: b, g)	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process: Principles of physicochemical and biological processes Lectures (individual) Case studies (group work), Project and problem-based teaching, The students are expected to read background material in the form of research articles and chapters from several handbooks available in digital form in the TU Delft library. Students can consult knowledge clips from online sub themes on specific topics. The assignments need to be carried out during the lecture period. Online lectures and on campus workshops Biological or chemical reactor performance analysis Simulations (for hypothesis formation) by student (group work) Groups contain 2/3 members.	

	Laboratory Experiment (group work) Data analysis and Modelling (group work) Presentation of results (group work) The students are expected to read background material in the form of research articles and chapters from several handbooks available in digital form in the TU Delft library. The assignments need to be completed during Q3.
Assessment	Students complete the module successfully when passing the 3 summative assessments: Principles of physicochemical and biological processes Individual written exam (50%) Biological or chemical reactor performance analysis phreeqC assignments (25%) Written Report on lab experiment (25%) A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Expected prior Knowledge	Completed MUDE and EE Stepping stone 1 and 2
Connection to other modules	This is a prerequisite module for B1 (in water technology track)
Academic Skills	Critical thinking, analytical thinking and problem solving are the main academic skills which will be trained. In addition, students will need to collaborate with their peers and thus organizational and interpersonal skills will be developed as well. Ethical reflections/considerations are included in a formative way.
Literature & Study Materials	The following books cover the main content of the module and relevant chapters will be offered to the students via Brightspace. Soil and water chemistry: an integrated approach. Michael E. Essington. CRC Press. ISBN 9781466573154 Water quality engineering Physical/chemical treatment processes. Mark Benjamin & Desmond Lawler. John Wiley & Sons. ISBN 978-1118169650 Biological Wastewater Treatment: Principles, Modeling and Design. Guanghai Chen, George A. Ekama, Mark C. M. van Loosdrecht, Damir Brdjanovic. IWA Publishing. ISBN (electronic): 9781789060362 Biological Wastewater Treatment: Principles, Modelling and Design: Examples & Exercises Additional background material and papers will be provided via BrightSpace.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Open book exam (calculator and paper-printed material allowed)
Collegerama	No

ENVM1501	Design and Modeling of Urban Water Infrastructure Systems	9
Responsible Instructor	Prof.dr. Z. Kapelan	
Instructor	Dr.ir. J.G. Langeveld	
Instructor	Prof.dr. G.J. Medema	
Instructor	Prof.dr.ir. J.P. van der Hoek	
Instructor	E.J.M. Blokker	
Instructor	D. Mehta	
Instructor	Dr. E. Abraham	
Instructor	Prof.dr. Z. Kapelan	
Contact Hours / Week x/x/x/x	14/x/x/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Urban water infrastructure systems sustain human life in towns and villages by supplying drinking water in sufficient quantity and quality, collecting and conveying wastewater and stormwater to a recipient whilst protecting human health, maintaining acceptable level of flood risk and protecting the natural environment. Modern urban water infrastructure is based increasingly on the concept of integrated water management ensuring that natural and other resources are (re)used in an efficient and sustainable manner. This module deals with concepts, systems, solutions, models, data and knowledge required for the effective design (and redesign / rehabilitation) of urban water infrastructure systems and their monitoring. A range of related issues are addressed covering water quantity, quality and other aspects. Both conventional, grid type systems (based on drinking water and sewer pipe networks) and latest off-grid type systems and solutions (such as sustainable drainage and other solutions) are considered including integrated water systems and solutions. Sub-theme "Design of urban water infrastructure systems" (link to learning objectives LO1, LO2 and LO3) This subtheme focusses on developing students understanding of concepts, system configurations, solutions, models, data and knowledge required for the effective design and redesign / rehabilitation of urban water infrastructure systems. Both drinking water and urban drainage systems are covered, including related modelling required for design. The focus of this subtheme is on water quantity related aspects of design. Sub-theme "Water quality processes in urban water infrastructure systems" (link to learning objectives LO3, LO4 and LO5). This subtheme focusses on developing a students understanding of water quality processes in urban water infrastructure systems and related monitoring. Water quality covers chemical parameters, microbiological parameters, physical parameters and the interaction between these parameters. Both drinking water and urban drainage systems are covered, including related water quality modelling.	
Study Goals	LO1: Design new and re-design existing urban water infrastructure systems. LO2: Design, dimension and analyse individual components of urban water infrastructure systems. LO3: Understand and apply relevant simulation and other models to analyse and plan new and suggest improvements to existing urban water infrastructure systems and their components. LO4: Design monitoring setup/networks for water quantity and water quality parameters taking specific challenges and dynamics of the range of urban water infrastructures into account. LO5: Study and understand water quality processes in urban water infrastructure systems using models and monitoring data and identify knowledge gaps in scientific literature on water quality processes.	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process: "Design of urban water infrastructure systems" This subtheme will be taught via lectures, assignment and demonstration of software tools. Lectures will be delivered in a conventional format together with suitable questions / discussions taking place during the lecture. The assignment will focus on the design of a real urban drainage or drinking water system with weekly feedback sessions provided to support the students. Relevant software tool demonstrations will be conducted too, via half-daily workshops. These include Epanet software for drinking water distribution systems and SWMM for urban drainage / sewer systems. At these workshops students will have to use these tools to answer in situ provided problems. "Water quality processes in urban water infrastructure systems" This subtheme will be taught via lectures assignments and field experiments. Lectures will be delivered in a conventional format together with suitable questions / discussions taking place during the lecture. The assignment considers the analysis and evaluation of water quality in a sewer system.	
Assessment	Students complete the module by passing the following: 1. Oral exam, single exam covering material taught in both sub-themes, 50% of the overall module grade. Design of urban water infrastructure systems: 2. Assignment report on urban water infrastructure design (drinking water or urban drainage related, 30%) Water quality processes in urban water infrastructure systems: 3. Assignment report on water quality processes in urban water infrastructure (20%) Note: All of above must be passed to complete the module. A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Expected prior Knowledge	Undergraduate hydraulics in civil engineering	
Connection to other modules	This module provides necessary input for the ENV1601 module.	
Academic Skills	Information collection and synthesis; Application of theory to situations in engineering practice; Making inferences of physical	

	situations by analysing observations/data; Use of modelling tools; Ethical reflection/consideration.
Literature & Study Materials	Design of urban water infrastructure systems Butler, D., Digman, C.J, Makropoulos, C. and Davies, J.W., Urban Drainage, Blackwells, ISBN-13: 978-1498750585 Savic, D. and Banyard, J., Water Distribution Systems, Thomas Telford Publishing, ISBN-13: 978-0727741127 Epanet software for the modelling of drinking water systems (both quantity and quality), https://www.epa.gov/water-research/epanet SWMM software, for the modelling of urban drainage / sewer system (both quantity and quality), https://www.epa.gov/water-research/storm-water-management-model-swmm Lecture notes (via Brightspace) Assignment manual with requirements and data for the design of an urban drainage system in part of Delft (via Brightspace) Papers and chapters from relevant literature (via Brightspace) Water quality processes in urban water infrastructure systems Hvitved-Jacobsen, T., Vollertsen, J., Nielsen, A.H. Sewer Processes. Microbial and Chemical Process Engineering of Sewer Networks. ISBN 9781439881774 Ashley, R.M., Bertrand-Krajewski, J.L., Hvitved-Jacobsen, T., Verbanck, M. Solids in Sewers. Characteristics, Effects and Control of Sewer Solids and Associated Pollutants. ISBN13: 9781900222914 Lecture notes (via Brightspace) Papers and chapters from relevant literature (via Brightspace) Assignment manual with requirements and data (via Brightspace)
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Calculator
Collegerama	Yes

ENVM1502	River Basin Hydrology and Water Management	9
Responsible Instructor	Dr. M. Hrachowitz	
Instructor	Dr. P.G. Ditmar	
Instructor	Dr.ir. R.J. van der Ent	
Instructor	Dr.ir. G.H.W. Schoups	
Instructor	Dr. M. Hrachowitz	
Instructor	Prof.dr.ir. N.C. van de Giesen	
Instructor	Dr.ir. R.W. Hut	
Instructor	Prof.dr.ir. R. Uijlenhoet	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Floods and droughts are among the most concerning water system threats and may be aggravated by land use and climate change. Water resources engineers use process understanding, mathematical models and observation technology including remote sensing to study floods, droughts and the effects of interventions. In this module students will learn about hydrological processes that occur at different scales in river basins and how they are affected by land use change, climate change, and/or human interventions. Focus lies on floods, droughts and the role of vegetation. Students will learn about the physics of hydrological processes and how to observe and model them and to evaluate the effects of management strategies on the river basins taking ethical considerations into account. Processes in river basin hydrology and water management Link to learning objectives of the module: A3: LO1. In this first topic students will focus on understanding the physical processes at different scales through lectures and discussing recent journal papers. The specific topics include: - Macroscale Laws, structure and organization in hydrological processes. - Role of Vegetation & land management in the Water Cycle - Flood processes & management - Cold region processes - Drought processes & management Remote sensing data and data assimilation in river basin hydrology and water management Link to learning objectives of the module: A3: LO3. In this second topic students will learn, via tutorials and assignments, about observation techniques, data handling and how to use them in modelling. - Remote sensing data types, sources and properties - Accessing, handling and processing data - Automatizing data access - Learning from data: basic concepts of data assimilation Models in river basin hydrology and water management Link to learning objectives of the module: A3: LO2. The third topic aims to conceptualize the hydrological processes such that the effects of climate and land use change can be quantitatively evaluated. Typical hydrological and river basins model concepts will be explained and how to calibrate and test them. - Model types and applications in Hydrology - Flow paths - Model parameter calibration and testing - Sources of uncertainty and the role of inverse, ill-posed problems - Model evaluation and selection techniques	
Study Goals	Floods and droughts are among the most concerning water system threats and may be aggravated by land use and climate change. Water resources engineers use process understanding, mathematical models and observation technology including remote sensing to study floods, droughts and the effects of interventions. In this module students will learn about hydrological processes that occur at different scales in river basins and how they are affected by land use change, climate change, and/or human interventions. Focus lies on floods, droughts and the role of vegetation. Students will learn about the physics of hydrological processes and how to observe and model them and to evaluate the effects of management strategies on the river basins taking ethical considerations into account. Processes in river basin hydrology and water management Link to learning objectives of the module: A3: LO1. In this first topic students will focus on understanding the physical processes at different scales through lectures and discussing recent journal papers. The specific topics include: - Macroscale Laws, structure and organization in hydrological processes. - Role of Vegetation & land management in the Water Cycle - Flood processes & management - Cold region processes - Drought processes & management Remote sensing data and data assimilation in river basin hydrology and water management Link to learning objectives of the module: A3: LO3. In this second topic students will learn, via tutorials and assignments, about observation techniques, data handling and how to use them in modelling.	

	- Remote sensing data types, sources and properties - Accessing, handling and processing data - Automatizing data access - Learning from data: basic concepts of data assimilation Models in river basin hydrology and water management Link to learning objectives of the module: A3: LO2. The third topic aims to conceptualize the hydrological processes such that the effects of climate and land use change can be quantitatively evaluated. Typical hydrological and river basins model concepts will be explained and how to calibrate and test them. - Model types and applications in Hydrology - Flow paths - Model parameter calibration and testing - Sources of uncertainty and the role of inverse, ill-posed problems - Model evaluation and selection techniques
Education Method	Teaching and learning that encourages students to take responsibility for their learning process: Processes in river basin hydrology and water management The teaching method is a combination of interactive lectures and exercises. Prior to lectures corresponding book chapters/handouts/papers should be studied. To engage students, short mini exercises are given during the lectures. Remote sensing data and data assimilation in river basin hydrology and water management After a short introduction, students follow tutorials to get hand-on experience with working with spatial data. Additionally, they work on weekly assignments. These assignments are part of the students portfolio. Models in river basin hydrology and water management The teaching method is a combination of interactive lectures and exercises. Prior to lectures corresponding book chapters/handouts/papers should be studied. To engage students, they work on assignments where they develop their own model concept including testing different calibration methods and assessing its performance.
Assessment	Students complete the module successfully when passing the 3 summative assessments: Processes in river basin hydrology and water management (35%): Written exam. Remote sensing data and data assimilation in river basin hydrology and water management (30%): The portfolio, consisting of the weekly assignments, will be graded. Models in river basin hydrology and water management (35%): Group report and presentation of project.
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Academic Skills	Knowledge of fundamental hydrological concepts and methods Analytical thinking; Application of theory to new situations using the relevant equations; Making inferences of physical situations by analysing observational data; Reading and discussing journal papers.
Literature & Study Materials	Processes in river basin hydrology and water management Material for the course will include (modified versions of): https://ocw.tudelft.nl/courses/hydrological-measurements/ (accessed 7-4-2021) https://ocw.tudelft.nl/courses/hydrology-of-catchments-rivers-and-deltas/ (accessed 7-4-2021) Selected papers Remote sensing data and data assimilation in river basin hydrology and water management Material for the course will include (modified versions of): https://ocw.tudelft.nl/courses/spatial-tools-in-water-resources-management/ (accessed 2-4-2021) Models in river basin hydrology and water management Material will be taken from publications on hydrological modelling including: Kirchner, J. W. (2006). Getting the right answers for the right reasons: Linking measurements, analyses, and models to advance the science of hydrology. <i>Water Resources Research</i>, 42(3). Beven, K. (2006). Searching for the Holy Grail of Scientific Hydrology: $Q = H(S? R?)$ A as closure. <i>Hydrology and Earth System Sciences Discussions</i>, 3 (3), 769-792. Hrachowitz, M., Savenije, H. H. G., Blöschl, G., McDonnell, J. J., Sivapalan, M., Pomeroy, J. W., ... & Cudennec, C. (2013). A decade of Predictions in Ungauged Basins (PUB) a review. <i>Hydrological sciences journal</i>, 58(6), 1198-1255. Hrachowitz, M., & Clark, M. P. (2017). HESS Opinions: The complementary merits of competing modelling philosophies in hydrology. <i>Hydrology and Earth System Sciences</i>, 21(8), 3953-3973. Blöschl, G., Bierkens, M. F., Chambel, A., Cudennec, C., Destouni, G., Fiori, A., ... & Renner, M. (2019). Twenty-three unsolved problems in hydrology (UPH) a community perspective. <i>Hydrological sciences journal</i>, 64(10), 1141-1158. This list will be updated each year with new and relevant publications.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	One self-written, one-sided A4 formula sheet (minimum font size 10pt)
Collegerama	Yes

ENVM1600	Water Treatment Technologies	15
Responsible Instructor	Prof.dr.ir. D. van Halem	
Instructor	Dr.ir. J.G. Langeveld	
Instructor	Prof. L.C. Rietveld	
Instructor	Prof.dr.ir. M.K. de Kreuk	
Instructor	Dr.ir. S.G.J. Heijman	
Instructor	Prof.dr.ir. D. van Halem	
Instructor	Dr.ir. H.L.F.M. Spanjers	
Instructor	Prof.dr.ir. J.P. van der Hoek	
Instructor	Prof.dr.ir. J.B. van Lier	
Instructor	K.M. Lompe	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Water treatment is needed to provide humans with water of sufficient quality to use it safely for purposes ranging from drinking to industry and to avoid pollution of the living environment. In module Water Treatment Technologies one will learn how to develop, research and design water treatment plants for the treatment of groundwater, surface water, municipal sewage and industrial water. Specific attention will be given to health impact, resource recovery and closing the urban water cycle. The module will cover separation, biological and chemical conversion technologies, including, membrane filtration, advanced oxidation, aerobic granular sludge and anaerobic treatment. In addition to project work and lectures, the module offers field visits to water treatment plants, experimental work in the laboratory and modelling exercises (Biowin, WAVE). The technologies covered in this module, include: Anaerobic digestion: slurry digestion and high-rate anaerobic treatment Aerobic sludge: flocculent and attached (granules) growth Biofiltration Pressure-driven membranes (reverse osmosis, nanofiltration, ultrafiltration) (Bio) sand filtration Ozone, chlorine, H2O2 and UV Electro(photo)chemical oxidation Activated carbon, GAC/ PAC/PACAS Ion exchange Resource recovery from sludge-based systems The course is organised with: Project (one can choose the "designer" or "researcher" track), assessed by group report, presentations and individual oral exam Lectures on water treatment technologies, assessed with written exam Laboratory, model, excursion and in-class assignments (formative assessments)	
Study Goals	Identify process conditions for drinking, industrial and sewage water treatment technologies based on the mechanistic understanding of coupled biological, chemical and physical processes. Assess the applicability of a water technology based on their effectiveness for target compounds, operational criteria, sustainability, and local conditions/ regulations. Design a water treatment plant including dimensions, hydraulic line, process scheme and control, considering budget, health impact, resource recovery and closing the urban water cycle. Research a water treatment technology, including hypothesis testing, experimental design, modeling and evaluating scientific literature. Evaluate project and research outcomes with the project team, clients and the scientific audience using communication and management tools.	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process: Project: Water treatment research and design Lectures (individual) Case studies (group work), Simulations if chosen by student (group work), Project and problem-based teaching, Excursions The students are expected to read background material in the form of research articles from the literature and chapters from several handbooks available in digital form in the TU Delft library. The assignments need to be carried out during the lecture period. Online lectures and on campus workshops Treatment Technologies Lectures (individual) Laboratory experiment and report (group work) Self-study of lecture and practice material provided on Brightspace Preparation of laboratory roadmap for practical Laboratory experiment on campus, Lectures online,	

Assessment	Students complete the module successfully when passing the 3 summative assessments: Written Exam: Water Treatment Technologies (6 ECTS) Group Presentation Project (1 ECTS) Oral Exam on Final Project Report (8 ECTS) Moments of formative assessments include: 2 or 3 Laboratory experiment reports Biowin and WAVE assignment Treatment train assignment Opening day challenge A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Academic Skills	Critical thinking; Analytical thinking; Problem solving; Collaborate with peers, Communicate results in presentations and project/scientific reports; Organizational and interpersonal skills; Ethical reflection/consideration.
Literature & Study Materials	The study material will be offered via Brightspace and consists of knowledge clips (with written hand-outs), scientific papers, conceptual process design lecture notes and Biowin handbook. Lectures will be available via opencourseware (OCW).
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations

ENVM1601	Operation, Control, Management and Adoption of Urban Water Infrastructure Systems	15
Responsible Instructor	Dr.ir. J.G. Langeveld	
Instructor	Dr.ir. J.G. Langeveld	
Instructor	Prof.dr. G.J. Medema	
Instructor	Prof.dr.ir. J.P. van der Hoek	
Instructor	Dr.ir. M.M. Rutten	
Instructor	E.J.M. Blokker	
Instructor	Dr. E. Abraham	
Instructor	Prof.dr. Z. Kapelan	
Instructor	Dr. R. Taormina	
Instructor	M.K. Mehta	
Contact Hours / Week x/x/x/x	24/x/x/x	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Urban water infrastructures must sustain urban living by supplying reliable drinking water, minimising contact risk with pathogens, maintaining urban flood risk at an acceptable level, and protecting the aquatic environment. On top of this, the infrastructures must contribute to circularity. This module deals with the operation, control, management and adaptation of urban water infrastructure systems, comprising both networks and off-grid systems. All the B-modules in the Water Resources Engineering track will have a workshop on ethical responsibility of engineers. Operations and control of urban water systems Link to learning objectives module B2: LO1, LO2 and LO7. Lectures about this topic will focus on operation of drainage systems, control of urban water infrastructures and data processing: Conventional operation of both drinking water and urban drainage systems Automated operation via real time control of urban water infrastructures. Cybersecurity of urban water infrastructure Data collection, storage, validation and mining for operation and maintenance of urban water infrastructures. Data sources are (internal) monitoring networks, call data, risks databases, insurance data and social media (especially for urban flooding). Sensors and sensor network design. Data validation. Asset management of urban water infrastructure systems Link to learning objectives module B2: LO3, LO4 and LO7. Lectures about this topic will focus on asset management of urban water infrastructure systems. Topics that will be covered are: Overview, inspection, deterioration, interventions and other related topics. Introducing AM concepts, highlight process steps, AM management methods, i.e., performance appraisal, financial accounting, portfolio modelling and management, and (technical) requirements and data needs per process step. Inspecting techniques for urban water infrastructures, related to the field of application/related failure mechanisms, both conventional and more advanced Deterioration modelling concepts ranging from deterministic models (impact of uneven soil subsidence on failure of rigid pipes) to cohort survival statistical models. Different methods for prediction of asset deterioration via pipe condition grades, burst rate, blockage and collapse rate will be shown with examples. Different types of interventions (from regular maintenance to rehabilitation and replacement). Deterioration modelling. Adaptation of urban water infrastructures Link to learning objectives module B2: LO5, LO6 and LO7. Lectures about this topic will focus on adaption to climate change, water demand management and urban pollution. Interaction with other infrastructure systems and related governance issues will also be addressed. Adaptation to climate change / uncertain future including urban flood risk management: Dealing with excess of water; basic definitions and concepts, flood risk assessment, different interventions and management options, urban pollution management, thermal energy recovery from piped systems, nature-based solutions Water demand management Emerging water quality issues and urban pollution management. Water quality impacts of emissions from urban water infrastructures: CSO, SSO, WWTP effluent. Entire range of impacts from acute toxic to long term build up. Including new topics: Emerging contaminants (include Coronavirus), microplastics Public health aspects related to urban water infrastructures, focussing on water supply (drinking water + alternatives) and wastewater and stormwater management (reducing contact risk with pathogens). Urban pollution management. Interactions with other infrastructure systems and related governance issues Physical interactions between water and wastewater, gas, road, telecom etc infrastructure; Material interactions: water-energy-food nexus	
Study Goals	B2-LO1: Design, evaluate and optimise control strategies for urban water infrastructures. (ILO: c) B2-LO2: Develop data driven management of urban water infrastructure systems using data mining and modelling technique. (ILO: e, f) B2-LO3: Apply asset management to urban water infrastructures. (ILO: c) B2-LO4: Study deterioration processes in urban water infrastructures using models and monitoring data and identify knowledge gaps in scientific literature on deterioration processes. (ILO: d, e, f, g, h) B2-LO5: Incorporate climate change adaptation and circularity in the design of urban water infrastructures. (ILO: c)	

Education Method	B2-LO6: Assess the impact of urban development/urban transition processes on urban water infrastructure. (ILO: b, h) B2-LO7: Conduct a design and a research project in a team and communicate results with relevant audience. (ILO: i, j, k) Teaching and learning that encourages students to take responsibility for their learning process: Operations and control of urban water systems Lectures 1 day workshop on cybersecurity Assignments excursion Asset management of urban water infrastructure systems Lectures Assignments Adaptation of urban water infrastructures Lectures Group research project Excursion
Assessment	Students complete the module successfully when passing the 4 summative assessments: Oral exam, single exam covering material taught in the three subthemes (50%). Operations and control of urban water systems (16,6%): Group design exercise Asset management of urban water infrastructure systems (16,6%): Group research report Adaptation of urban water infrastructures (16,6%): Group research report A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Expected prior Knowledge	A2 module
Academic Skills	Information collection and synthesis; Application of theory to new situations using the relevant equations; Making inferences of physical situations by analysing observations/data; Ethical reflection/consideration.
Literature & Study Materials	Operations and control of urban water systems Lecture notes and papers/chapters from relevant literature will be provided via BrightSpace. The list of papers will annually be updated and will contain papers like: Langeveld, J. G., L. Benedetti, J. J. M. de Klein, I. Nopens, Y. Amerlinck, A. van Nieuwenhuijzen, T. Flameling, O. van Zanten, and S. Weijers. 2013. "Impact-Based Integrated Real-Time Control for Improvement of the Dommel River Water Quality." <i>Urban Water Journal</i> 10 (5): 312-329 van Daal P., Gruber G., Langeveld J., Muschalla D., Clemens F. (2017). Performance evaluation of real time control in urban wastewater systems in practice: Review and perspective. García, L., Barreiro-Gomez, J., Escobar, E., Tellez, D., Quijano, N., Ocampo-Martinez, C., 2015. Modeling and real-time control of urban drainage systems: a review. <i>Adv. Water Resour.</i> 85, 120e132. http://dx.doi.org/10.1016/j.advwatres.2015.08.007. Asset management of urban water infrastructure systems Background material, papers and chapters from relevant literature will be provided via BrightSpace. The list of papers will annually be updated and will contain papers like: Tscheikner-Gratl, F., Caradot, N., Cherqui, F., Leitão, J.P., Ahmadi, M., Langeveld, J.G., Le Gat, Y., Scholten, L., Roghani, B., Rodríguez, J.P., Lepot, M., Stegeman, B., Heinrichsen, A., Kropp, I., Kerres, K., do Céu Almeida, M., Bach, P.M., Moy de Vitry, M., Sá Marques, A., Simões, N.E., Rouault, P., Hernandez, N., Torres, A., Wery, C., Rulleau B., & Clemens, F. (2020). Sewer asset management state of the art and research needs. <i>Urban Water Journal</i>. Dirksen, J., Clemens, F. H. L. R., Korving, H., Cherqui, F., Le Gauffre, P., Ertl, T., Snaterse, C. T. M. (2013). The consistency of visual sewer inspection data. <i>Structure and Infrastructure Engineering</i>, 9(3), 214-228. van Riel, W., Langeveld, J.G., Herder, P.M., Clemens, F.H.L.R.. (2014). Intuition and information in decision-making for sewer asset management. <i>Urban Water Journal</i>, 11 (6), pp. 506-518. Adaptation of urban water infrastructures Background material, papers, chapters from relevant literature and an excursion report will be provided via BrightSpace. The list of papers will annually be updated and will contain papers like: Marlow, D. R., Moglia, M., Cook, S., Beale, D. J., Land, C., & Road, G. (2013). Towards sustainable urban water management : A critical reassessment. <i>Water Research</i>, 47(20),71507161. https://doi.org/10.1016/j.watres.2013.07.046 Nieuwenhuis, E., Cuppen, E., Langeveld, J., & De Bruijn, H. (2020). Towards the integrated management of urban water systems. <i>Journal of Cleaner Production</i> Wong, T.H.F. , Rogers B.C. Brown, R.R. 2020. Transforming Cities through Water-Sensitive Principles and Practices. <i>One Earth</i>. 3 (4), pp 436-447
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	none

ENVM1602	Regional Hydrology	15
Responsible Instructor	Dr. T.A. Bogaard	
Instructor	Prof.dr.ir. M. Bakker	
Instructor	Dr. T.A. Bogaard	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Our regional water systems are far from natural anymore, with human-made or modified rivers, canals, and subsurface water systems. This infrastructure cannot be seen separately from its surrounding environment, while keeping the relation with the desired functionalities. In this module students will learn about open water systems, with their canals (human-made or modified) and structures, as well as the subsurface water system. Focus will be on the hydraulics of the water infrastructure (flows), but issues of water quality, ecology and morphology are included as well. Project: Water systems Link to the learning objectives of the module: B3: LO1, LO2, LO5 and LO8. Students learn how to connect the individual behaviors of infrastructural elements in a water system, how to model water systems, using available software, and how to define goals of selected (types of) systems in terms of desired water flows (like drainage-irrigation, flood-drought, urban-rural) by focusing on the following topics: Existing water systems are analyzed, using several sessions specifying quantification of behavior - allowing linking to the other sub themes. Exchanges between teams (peer reviews) and students working on different modelling approaches allow students to develop insight into model use and system behavior. Specific problems in selected water systems are addressed, and students propose measures relating explicitly to the other sub themes. All the B-modules in the Water Resources Engineering track will have a workshop on ethical responsibility of engineers. Groundwater modelling Link to learning objectives of the module: B3: LO2 and LO4. In this unit, students learn how to model groundwater systems, by quantifying groundwater flow and heads. Different model approaches are applied in exercises to assess the environmental impact of man-made changes in the geohydrological system. Specific topics include: The analytic element method and the finite difference method (including MODFLOW). Comparison of grid-based methods and analytical methods. Design of well fields including capture zones and wellhead protection Pumping test analysis Travel times Time series analysis of groundwater time series. Prediction of groundwater heads in wet and dry periods. Environmental (eco)hydrology Link to learning objectives of the module: B3: LO2 and LO3. This sub theme focuses on quantifying and measuring the regional water system in its dynamic surrounding environment. Hands-on measuring techniques will be applied in a field setting, where after students critically analyse and reflect on the collected data. Next to the focus on quantifying the fluxes, special attention will be paid on water quality, ecology and geomorphology. Specific topics include: Geomorphological processes Measuring e.g., precipitation, evaporation, interception, soil composition, infiltration, hydraulic conductivity, piezometry Applying river discharge measurements by different methods Observation of water quality, ecosystem characteristics and geomorphology Understand the influence of discharge characteristics on water quality, ecosystem characteristics and geomorphology Tracer hydrology	
Study Goals	B3-LO1: Define how a water systems shape relates to the function(s) of the water system, in terms of (surface and subsurface) water fluxes. (ILO: b, h, l) B3-LO2: Quantify the water fluxes in selected systems, including non-steady states and interactions between fluxes. (ILO: a, f) B3-LO3: Use of measurement methodologies for hydrology in its dynamic biotic and abiotic environment and critically interpret the obtained data. (ILO: a, e, g) B3-LO4: Develop mathematical models to explain groundwater flow and heads to predict environmental impact. (ILO: a, f, g) B3-LO5: Propose measures to realize desired functionalities in water systems, considering short- and long-term criteria. (ILO: c, d, h, l) B3-LO6: Conduct a project in a team and communicate results with relevant audience. (ILO: i, j, k)	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process: Project: Water systems The sub-theme builds on a series of workshops, with focus areas per week, that relate to the project a team works on. Project teams meet at least once every week with their supervisor to discuss progress.	

Assessment	Groundwater modelling After the lectures, students will work in Jupyter notebooks to practice with the theory. In each notebook students will work on a model assignment. The notebooks are part of one's portfolio of the lecture series. Environmental (eco)hydrology The teaching method consists of a fieldwork, where students will get hands-on experience with carrying out hydrological fieldwork. The collected field data has to be screened and analysed. At the end of the day, one will present the results. Additionally, lectures on the theory of morphodynamics and ecology will be provided. During these lectures students will study scientific papers on the interaction of (a)biotic and hydrological systems, that will be discussed in class. Students complete the module successfully when passing the 3 summative assessments: Project: Water systems (40%): Assessment 1: Final exam testing quantification of water systems, with at least 30% of the questions related to assignments prepared during the course. Assessment 2: Team report, including details on individual contributions, plus reflection per team member on personal skills: teamwork, reporting, communication. Groundwater modelling (40%): The weekly assignments that are part of the student's portfolio are graded. Environmental (eco)hydrology (20%): Written exam.
Academic Skills	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment. Analytical thinking; application of theory to new situations using the relevant equations; making inferences of physical situations by analysing observational data; reading and discussing journal papers; Ethical reflection/consideration.
Literature & Study Materials	Project: Water systems Building on the study material from the Stepping Stone I and II Modules (which cover water design formula for example), additional study material is provided in the shape of guided assignments, with a problem description plus two or three papers relevant for the topic at hand. Per project, a starting kit will be collected, including a project description, key data, three relevant papers, and a list of stakeholders. During the project work, the teams collect additional material, in close contact with the supervisor.
Judgement	Groundwater modelling Weekly Jupyter Notebooks, with a model assignment, will become available. Environmental (eco)hydrology Part of field course syllabus CIE5471 Hydrological and Ecological Fieldwork in River Systems; List of scientific articles, book chapters, hand-outs and powerpoint presentations (see CIE5580 Ecology and Morphodynamics in Catchments) For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations

ENVM1603	Water Resources Engineering and Management	15
Responsible Instructor	Dr.ir. M.W. Ertsen	
Instructor	Dr.ir. M.W. Ertsen	
Instructor	Dr. E. Mostert	
Instructor	Dr. R.R.P. van Nooijen	
Instructor	Dr.ir. O.A.C. Hoes	
Instructor	Dr. S. Pande	
Instructor	Dr. E. Abraham	
Contact Hours / Week x/x/x/x	28/x/x/x	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This module focuses on water systems infrastructures of different types. Regional, rivers and subsurface water systems are included. The module combines analysing water systems with finding options for their (re)design and operation. Project: Water systems Link to the learning objectives of the module: B4: LO1, LO2 and LO7. Students learn how to connect the individual behaviour(s) of infrastructural elements in a water system, how to model water systems, using available software, and how to define goals of selected (types of) systems in terms of desired water flows (like drainage-irrigation, flood-drought, urban-rural) by focusing on the following topics: Existing water systems are analysed, using several sessions specifying quantification of behaviour - allowing linking to the other sub themes. Exchanges between teams (peer review) and students working on different modelling approaches allow students to develop insight into model use and system behaviour. Specific problems in selected water systems are addressed, and students propose measures relating explicitly to the other sub themes. Integrated Assessment of Water Systems Link to the learning objectives of the module: B4: LO3, LO4 and LO7. The changing climate and socioeconomic conditions will likely challenge the functioning of current water systems. Assessing such uncertainties requires understanding how the functioning of systems is valued in the short and long run. Such understanding allows assessing diverse water engineering options that are available at present. Based on the observation and predictions of effects of engineering solutions, this subtheme develops creative interventions for sustainable water futures (e.g., in context of SDGs) through an interdisciplinary approach for designing and implementing such solutions by addressing the following topics: Design and simulation of relevant climate and socioeconomic scenarios Economic and non-market valuation of system functioning in the short and long run based on the scenarios. Analysis and proposal of engineering solutions with functionalities in water systems that are climate resilient and economically viable. Control of Water Systems Link to the learning objectives of the module: B4: LO5, LO6 and LO7. In a rapidly changing world, where the demands of society approach or exceed the limits imposed by natural resources, engineering solutions need to make efficient use of resources and quickly adapt to change. This sub theme teaches how to derive indicators and design a control algorithm for a water system by focussing on the following topics: Rapid adaptation requiring indicators of the desired functioning of the system that can be calculated frequently and automatically. Semi-automated set of rules (algorithms) to steer indicators to their predefined reference values.	
Study Goals	All the B-modules in the Water Resources Engineering track will have a workshop on ethical responsibility of engineers. B4-LO1: Define how a water systems shape relates to the function(s) of the water system, in terms of (surface and subsurface) water fluxes. (ILO: a, b, c, e, f) B4-LO2: Quantify the water fluxes in selected systems, including non-steady states and interactions between fluxes. (ILO: a, b, c, d, e, f, g) B4-LO3: Assess engineering options for water systems in terms of societal benefits, environmental impact, livelihoods, and adaptive capacity. (ILO: a, c, f, h, l) B4-LO4: Assess an economic analysis of these engineering options in terms of current and future investments, costs and benefits. (ILO: a, c, f, h, l) B4-LO5: Propose measures to realize desired functionalities in water systems, considering short- and long-term criteria. (ILO: a, b, c, d, e, f, g, h, l) B4-LO6: Determine how a proposed set of measures to realize desired functionalities may be implemented. (ILO: b, d, e, g, h, l) B4-LO7: Design and implementation of a proposed set of measures to realize the desired system performance. (ILO: a, b, c, e, f)	
Education Method	Project: Water systems The sub-theme builds on a series of workshops, with focus areas per week, that relate to the project a team works on. Project teams meet at least once every week with their supervisor to discuss progress. Integrated Assessment of Water Systems Problem based learning, select group/individual assignments (multiple, linked to specific case studies/problems (e.g., investing in geothermal energy storage, water for food and environment quality nexus at basin scale), with assignments per session. The analysis builds on relevant MUDE elements and the EE programme module. Control of Water Systems Problem based learning, select group/individual assignments, supported by some specific lectures. Building on elements that have been covered in the MUDE, the sub theme introduces control issues and options, plus the relevant underlying mathematics	

Assessment	Students complete the module successfully when passing the 4 summative assessments, related to the sub-theme's as indicated below. Project: Water systems (40%): Assessment 1: Final exam testing quantification of water systems, with at least 30% of the questions related to assignments prepared during the course. (25%) Assessment 2: Team report, including details on individual contributions, plus reflection per team member on personal skills: teamwork, reporting, communication. (25%) Integrated Assessment of Water Systems (30%): Assessment 3: Written exam, with at least 30% of the questions related to assignments prepared during the course. Control of Water Systems (30%): Assessment 4: Written exam, with at least 30% of the questions related to assignments prepared during the course
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment. Steady flow in surface water flows in canals and structures; basic aspects of dynamics in water systems; basic aspects of hydrology; basic aspects of societal aspects of water systems; mathematics
Connection to other modules	This module can be selected from each A module in the Water track of the MSc Environmental Engineering. This module can be selected from each A module in the tracks Hydraulic Engineering and Hydraulic&Offshore Structures of the MSc Civil Engineering.
Academic Skills	Problem analysis; Data application; Quantification; Integration; Writing; Communication; Teamwork; Ethical reflection/consideration.
Literature & Study Materials	Project: Water systems Building on the study material from the Stepping Stone I and II Modules (which cover water design formula for example), additional study material is provided in the shape of guided assignments, with a problem description plus two or three papers relevant for the topic at hand. Per project, a starting kit will be collected, including a project description, key data, three relevant papers, and a list of stakeholders. During the project work, the teams collect additional material, in close contact with the supervisor. Integrated Assessment of Water Systems Study material is provided in the shape of guided assignments, with a problem description plus two or three papers relevant for the topic at hand. As hydro-economic modelling is a key element of this module content, Roobavannan et al., 2020. Sustainability of agricultural basin development under uncertain future climate and economic conditions: socio-hydrological analysis. Ecological Economics, 174, 106665. (in press) is foreseen as study document. Further specific content depends on what is covered in the MUDE and EE programme module. Control of Water Systems Study material is provided in the shape of guided assignments, with a problem description plus two or three papers relevant for the topic at hand. One overarching study document is envisioned, but the decision to use an existing document (book) or compose a (more limited) sub-theme specific one will be made later, as the content depends on MUDE elements.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	TBD
Collegerama	No

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE Track Resource & Waste Engineering (R&WE)

ENVM1100	Fundamentals of RW and E	15
Responsible Instructor	Dr. H.M. Jonkers	
Instructor	Dr. H.M. Jonkers	
Instructor	Dr.ir. M. Ottele	
Instructor	Dr. M.W.N. Buxton	
Instructor	Dr. P.J. Vardon	
Instructor	Dr.ir. M.C.M. Bakker	
Instructor	Prof.dr.ir. T.J. Heimovaara	
Instructor	Dr. D.F.J. Schraven	
Instructor	Dr. J. Gebert	
Instructor	Dr.ir. A.A.M. Dieudonné	
Instructor	Dr. B.M. van Breukelen	
Instructor	A. Soleymani	
Practical Coordinator	Dr. F. Di Maio	
Contact Hours / Week x/x/x/x	x/6/6/x	
Education Period	2 3	
Start Education	2	
Exam Period	2 3 4	
Course Language	English	
Course Contents	The stepping stone Fundamentals of Resource & Waste Engineering is compulsory for all students following the Resources and Waste Engineering (R&WE) track. This module provides an introduction to three scientific fields, which will be specifically addressed in the three respective themes: (1) Circularity concepts in resource and waste engineering, aiming for reduction of use of primary resources and emission of harmful components to the soil-, water-, and atmosphere compartments; (2) Environmental impacts of resource production and use and associated production of waste, with focus on primary and secondary resources typically produced for and used in the geo- and civil engineering building sector; (3) Biogeochemical processes occurring in the soil-, water-, and atmosphere compartments, with specific focus on how biogeochemical processes transform specific geo- and building sector related contaminants in these compartments and what the consequences are for the environmental quality of these compartments. Goal: The objective of the Resource & Waste Engineering stepping stone is to gain conceptual knowledge and understanding of environmental problems arising from geo- and civil engineering practices and how circular building and waste management strategies and (stimulation of) specific biogeochemical processes can mitigate these negative effects. The module content is organized in the following three themes: (1) Circularity in resource and waste engineering Link to learning objectives of the module: SS: LO1, LO2, LO3, LO5, LO6 and LO7. Central topic of this theme is the assessment of the theoretical background of resource efficiency principles and circular resource and waste concepts including value creation from waste and development of circular business models. The obtained knowledge will provide insight on how circular resource and waste management practices can be used to support the economy and achieve political goals for reducing use of primary resources and emissions of harmful components to the environment. This theme particularly focuses on the following topics: Concept, history, scope and reach of the circular (building) economy; General principles of circularity and resource efficiency; Implications and implementation of circularity and circular economy in the context of supply chains and supply; networks; Value creation from secondary resources; Sustainable and circular business models; Legislative framework addressing circularity and resource efficiency; Quantification of circularity performance through LCA methods. This theme builds further on prior knowledge obtained in the ENV programme module and provides theoretical background knowledge as a basis for further deepening and elaboration on specific circular resource and waste management techniques, cases and applications in subsequent R&WE track A and B modules. (2) Environmental impacts of resource utilisation Link to learning objectives of the module: SS: LO1, LO5 and LO7. Main aim of this theme is gaining understanding of the solid waste flows associated to mining- and civil engineering building practices and how these are linked to the production, use and end-of-life of primary and secondary resources and the environmental effects they cause in the soil-, water- and atmosphere compartments. This theme particularly focuses on the following subjects: Introduction to types, properties and quantities of the main construction- and mining-related solid waste sources and streams; Environmental effects (impacts) and hazard risks of construction- and mining-related solid waste sources and streams; Legislative frameworks and ethical aspects of construction- and mining-related waste; Solid waste recycling- and disposal options in relation to material and energy flows Introduction to technical routes for converting- and valorising waste; Evaluating environmental effects and hotspots occurring in the exploration-, extraction- and production stage of mineral resources; The theory of physical and chemical transport phenomena Quantitative assessment of source emission and its impact on objects; Environmental impacts associated to geothermal energy production; Introduction to (overview of) possible remediation measures of environmental impacts related to solid waste production and processing; Ethical aspects of resource utilization will also be discussed in an interactive lecture organized together with the Faculty of Technology, Policy and Management. This theme builds further on prior knowledge obtained in the ENV programme module and provides theoretical background knowledge as a basis for further deepening and elaboration on remediation measures of environmental impacts of geo- and civil engineering related waste production and processing in subsequent R&WE track A and B modules.	
Course Contents Continuation		

	(3) Effect of biogeochemical processes on environmental quality Link to learning objectives of the module: SS: LO4 and LO7. This theme builds further on the introduction to environmental quality provided in the Environmental Engineering core programme module. The main aim of this theme is gaining understanding of how biogeochemical processes occurring in soil, water and atmosphere are involved in the conversion of specific contaminants and what consequences are for the environmental quality of these three compartments. Theories are explained and calculations are done and applied to various cases. This theme particularly focuses on the following subjects: Composition and structure of soil Hydraulic and mechanical behaviour of soil Retention and reactivity of soils sorption of organics, ions, metals Transfer of gasses between soil, water and air Alkalinity, pH-buffering, and Acid / Base equilibria Dissolution and precipitation of minerals, calculation of activity coefficients Basics of chemical modelling (PHREEQC): speciation and water-mineral interaction Biogeochemical reactions (Gibbs free energy, Standard Potential, van t Hoff equation) Microbial conversion of organic and inorganic contaminants Basics of redox reactions including calculation of oxidation states and chemical oxygen demand plus introduction on the occurrence of the main redox processes in groundwater and lakes This theme provides theoretical background knowledge as a basis for further deepening and elaboration on contaminant remediation measures of geo- and civil engineering related waste production and processing in subsequent R&WE track A and B modules.
Study Goals	On the successful completion of this module, the students are able to: SS-LO1: Identify the possibilities and limitations of resource circularity and solid waste management options and have knowledge on (critical) resource and waste flows. (ILO: g) SS-LO2: Create value from waste, and specifically implement concepts that enable circular principles to be applied to resource and waste streams relevant for the geo- and civil engineering building sector. (ILO: l) SS-LO3: Recognise environmental challenges and economic opportunities in solid waste management and translate these into scientific research questions and/or engineering opportunities. (ILO: d) SS-LO4: Develop numerical models and explain and verify, using scientific methods, the underlying biogeochemical processes that affect the environmental quality of soil, water and air. (ILO: a, f) SS-LO5: Take into account the national and EU goals and regulations when developing engineering solutions for waste management options, including collection, separation, treatment, transport, re-use, recovery, recycling and final disposal. (ILO: h) SS-LO6: Quantify circularity and environmental performance of materials (resources and wastes) and building processes relevant for the geo- and civil engineering sector using Life Cycle Assessment (LCA) tools. (ILO: g) SS-LO7: Communicate the essential principles of resource and waste engineering both orally and in writing. (ILO: i)
Education Method	Form of instruction per theme is the same: Class lectures and workshops of in total 6 times 45 minutes per week for a total of 8 weeks will be the primary mode of instruction to deliver the theoretical concepts of this theme. Background information in form of research articles and chapters from textbooks will be provided to the students in digital form via the TU Delft library. A group case study assignment will be given to students comprising a literature study on a specific geo- or civil engineering related waste stream and its associated environmental effects, reporting of findings in a written report, and presentation of results to fellow students. The assignment needs to be carried out and finalized within the lecture period.
Assessment	Obtained theoretical knowledge will be assessed through a (written) digital (Möbius) individual exam (50% of final grade) and literature review-, reporting- and presentation skills through an individual project portfolio (30% of final grade) and one group case study assignment (20% of final grade). An intermediate self-reflection and fellow student peer feedback on performance of individual students to the group process is part of the group case study assignment.
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Academic Skills	Knowledge obtained in preceding and parallel running Core Program Module ENVM1000 Principles of Environmental Engineering
Literature & Study Materials	Analytical assessment, reporting- and communication skills are the core competencies that will be gained in this stepping stone module. In addition to individual assignments students will also gain experience in collaborating in groups and learn how to effectively communicate obtained results. Project planning, organizational management-, reporting- and presentation skills will also be gained at individual level. Lecture slides, papers, and chapters from relevant literature will be provided via Brightspace. Some examples per theme include: (1) Circularity in resource and waste engineering Platform CB'23 - Measuring circularity (2019) Core method for measuring circularity in the construction sector MAJ Huijbregts et al (2017) ReCiPe 2016: a harmonized life cycle assessment impact method at midpoint and endpoint level. Int J Life Cycle Assess 22: 138-147 (2) Environmental impacts of resource utilisation Zhengfu Bian et al. (2012) The Challenges of Reusing Mining and Mineral-Processing Wastes. Science 337: 702-703 DM Franks et al. (2011) Sustainable development principles for the disposal of mining and mineral processing wastes. Resources Policy 36 (2): 114-122 G Lopez et al (2017) Thermochemical routes for the valorization of waste polyolefinic plastics to produce fuels and chemicals. A review. Renewable and Sustainable Energy Reviews 73: 346-368 (3) Effect of bio-geo-chemical processes on environmental quality Bertron (2014) Understanding interactions between cementitious materials and microorganisms: a key to sustainable and safe concrete structures in various contexts. Materials and Structures 47: 1787-1806 DA Nimick et al (2011) Diel biogeochemical processes and their effect on the aqueous chemistry of streams: A review. Chemical Geology 283: 317 K Soetaert et al (2007) The effect of biogeochemical processes on pH. Marine Chemistry 105: 30-51
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (R&G): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Dictionary
Collegerama	Yes

ENVM1200	Resource Engineering	9
Responsible Instructor	A. Varveri	
Instructor	Dr. F. Di Maio	
Instructor	Prof.dr.ir. J.W.G. van de Kuilen	
Instructor	Dr. H.M. Jonkers	
Instructor	Dr.ir. M. Ottele	
Instructor	A. Varveri	
Instructor	Dr.ir. M.C.M. Bakker	
Instructor	Dr. J. Gebert	
Instructor	Prof. M. Veljkovic	
Instructor	Dr. Y. Mosleh	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The R&WE track will start with only one module A. This module A on Resource Engineering is therefore compulsory for all students following the Resources and Waste Engineering (R&WE) track. The module builds further on the topics addressed in the R&WE stepping stone and gives students an introduction to the flows of resources and wastes, as well as to the types and properties of waste that result from human activity. Students will learn about using methodologies to quantify changes in resource demand and evaluate these in terms of their environmental implications and resource availability (critical materials). The module will further explore the issues and variables that play a role in using secondary resources more efficiently, and in closed loops. In this context, the limitations for circularity and the role of bio-based solutions (renewable and bio-based materials) in achieving circularity will also be addressed. Module A offers an engineering perspective to address key aspects related to circular production and construction processes and introduces a range of engineering tools to assess environmental performance. Goal: The objective of the Resource Engineering module is to gain in-depth knowledge of managing material flows and waste streams in closed loops and learn how circular and bio-based economy approaches can be implemented in production and construction processes, thus contributing to increasing the use of renewable and recyclable resources while minimizing the consumption of energy and raw materials. The module content is organized in the following two themes: (1) Flows of materials, resources and residual wastes (Link to learning objectives of the module: A-LO1, A-LO2, A-LO3, A-LO7 and A-LO8) Central topic of this theme is the analysis and management of material flows and waste based on circularity and bio-economy knowledge, with the aim of reducing environmental impacts. (2) Circular construction processes and sustainable materials (Link to learning objectives of the module: A-LO3, A-LO4, A-LO5, A-LO6 and A-LO7) Central topic of the theme is the application of circularity and bio-economy concepts on construction processes and materials and their evaluation based on tools for the quantification of the environmental performance at system and material levels.	
Study Goals	Upon successful completion of this module, the students are able to: A-LO1: Explain principles, methodologies and concepts related to waste management, circularity and construction materials. (ILO: a) A-LO2: Evaluate waste management options, considering the possibilities and limitations of resource circularity. (ILO: a, c, d, l) A-LO3: Assess technologies for separation, sorting of and resource recovery from solid waste by implementing concepts that enable material circularity. (ILO: a) A-LO4: Evaluate the state-of-the-art circular and bio-based economy approaches and methodologies for production and construction processes. (ILO: c, i, l) A-LO5: Evaluate the environmental impacts and performance of production and construction activities by applying both quantitative and qualitative methodologies. (ILO: b, d, g) A-LO6: Develop engineering solutions for implementing circularity and bio-based concepts in production and construction processes based on environmental, economic, societal and legislative grounds. (ILO: c, d, h, l) A-LO7: Demonstrate the capacity to effectively communicate (both written and oral) and share results and opinions with peers, engineers, and academics. (ILO: i) A-LO8: Provide constructive feedback, managing conflicts in diverse teams, and tackle together challenges in implementing circularity and bio-based economy. (ILO: j, k).	
Education Method	(1) Flows of materials, resources and residual wastes - Interactive lectures (on-campus or online) will be the primary mode of instruction to convey the theoretical concepts of the module. - Research articles from the literature and chapters from several handbooks available in digital form in the TU Delft library will be provided to the students as background material. - An assignment will be given to students in groups (of 2 persons) to formulate solutions for management of waste (household, industrial, agricultural, etc.) and allow students to learn from each other. The assignments need to be carried out during the lecture period. (2) Circular construction processes and sustainable materials - Interactive lectures (on-campus or online) will be given to teach the theory part and provide specific knowledge and discuss the tools and methodologies for determining and improving circularity and sustainability in production and construction processes. - A real-life case study (project) will be presented to the students (in groups of 2-3 persons) early in the theme. The aim of the group project is to quantify and evaluate the environmental performance of a product/process/structure and propose engineering solutions to improve it.	
Assessment	To successfully complete the module, students must pass the three (3) summative assessments as described below: (1) Module level	

	The knowledge of theory and problem-solving skills are tested by a written exam (30%). (2) Flows of materials, resources and residual wastes The analysis skills are tested through a group assignment (20%). (3) Circular construction processes and sustainable materials The design, writing and presentation skills are assessed through a group project (50%). The minimum grade for each partial assessment is 5.8. In this module, the group processes are also assessed. Except for the group assignment and the final group report, the students have to individually provide a peer review (the template will be provided by the instructor) on the individual contributions of their group members and a self-reflection on what they have learned and the dynamics of their group. A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Tags	Design Group work Project Small groups Sustainability
Connection to other modules	Module A consolidates and builds on prior knowledge obtained in the ENV programme module and R&WE stepping stone themes, and prepares for more advanced topics in the R&WE B-modules.
Academic Skills	Collection and synthesis of information, critical thinking, analytical thinking and problem solving are the main academic skills which will be trained. In addition, students will need to collaborate with their colleagues, produce a group report and present their work, so organizational, interpersonal, writing and presentation skills will be developed as well.
Literature & Study Materials	Lecture slides, papers, and chapters from relevant literature will be provided via Brightspace. Some examples include: (1) Flows of materials, resources and residual wastes M. Agnoletti, S. N. Serner (editors), Ch7: History of Waste Management and the Social and Cultural Representations of Waste, The Basic Environmental History, 2014 G. Tchobanoglous, H. Theisen, S. Vigil, Integrated Solid Waste Management, McGraw-Hill. (2) Circular construction processes and sustainable materials Sustainable steel Buildings, ed. B.Hauke, M. Kuhnhenne, M. Lawson, M.Veljkovic, 2016, p.360, Wiley. Mantalovas, K.; Di Mino, G.; Jimenez Del Barco Carrion, A.; Keijzer, E.; Kalman, B.; Parry, T.; Lo Presti, D. European National Road Authorities and Circular Economy: An Insight into Their Approaches. Sustainability 2020, 12, 7160. F. Di Maio et al, Measuring resource efficiency and circular economy: A market value approach, Res, Cons. & Rec, 2017 (122), pp. 163171. R., U., Ayres, Economic growth: politically necessary but not environmentally friendly, Ecological Economics 15 (1995) 97-99 United Nations Environment Programme, 2014. Sand, rarer than one thinks. Environ. Dev. 11, 208218. Gebremariam et al., Comprehensive study on the most sustainable concrete design made of recycled concrete, glass, and mineral wool from C&D wastes; Construction and Building Materials.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations

ENVM1300	Waste Processing Technologies	15
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Instructor	A. Varveri	
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Instructor	Dr.ir. M.C.M. Bakker	
Instructor	Dr. D.F.J. Schraven	
Instructor	Dr. M. Soleymani Shishvan	
Instructor	Dr. J. Gebert	
Instructor	A.T. Gebremariam	
Contact Hours / Week x/x/x/x	x/x/x/5-10 (exact number of contact hours is not clear, but ranges between 5 and 10 hours)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Module B1 focuses on routes and technologies that produce high-quality raw materials from different solid waste sources with high economic and societal impacts and a degree of urgency. Examples are household waste, WEEE, plastics, metals, industrial wastes, incineration residues, and tailings from mines. Routes are designed to uphold and further develop the circular economy's generic principles and concepts as taught in the stepping stone and Module A: i.e., effective solutions for recovery of materials from End-of-Life constructions, design out harmful residues and emissions to water, air, and soil, and prevent the depletion of non-renewable resources. This module will further explore the issues and variables that play a role in preparing and using secondary resources more efficiently, safeguarding the environment, and in closed loops. In this context, the limitations of circularity will also be addressed. The student will be challenged to apply the knowledge and skills provided by the expert teaching teams in a final project of design and analysis (B1-3), in compliance with regulations and discussing ethical aspects. This module also addresses the development of soft skills such as presentation techniques, professional reporting, team working, and efficient communication with internal and external experts. Goal: Gain knowledge and skills to understand materials and waste flows and to develop technologies for recycling, resource recovery, waste separation, characterisation, and disposal in the context of the circular economy. The module content is organised in the following three themes: (1) Resources recovery from End-of-Life constructions Link to learning objectives of the module: B1: LO1, LO2, LO3, LO5, LO8, and LO10. With a focus on the environment and the economy, this theme offers an engineering perspective to address critical aspects related to circular production and construction processes, advanced technologies for recycling, and recovery of resources from construction and demolition waste in a circular economy. The following topics will be addressed: - Regulations - Processes and procedures - Material labels - Waste flows - Dismantling strategy - Demolition techniques - Collection strategies - Primary and secondary raw materials - Innovative technological solutions that foster circularity - Routes to market - Environmental aspects (2) Waste collection and separation techniques Link to learning objectives of the module: B1: LO3, LO4, LO5, LO6, LO7 and LO8. Knowledge of solid waste separation principles and techniques is key to efficient recycling and improving the sustainability of material resources. This theme provides detailed knowledge about the flows of resources and wastes resulting from human activities. This theme will further explore the issues and variables that play a role in using secondary resources more efficiently and in closed loops. In this context, the practical limitations of circularity will also be addressed. The theme concentrates on the different steps and processes comprised in the solid waste management chain, from waste collection to treatment, recycling, and disposal. Taking into account the issues of sustainability and circularity, the theme provides knowledge on the processes costs, the quality, and the value of the secondary materials resulting from the different processes/treatments discussed. The following topics will be addressed: - Collection methods - Physical waste properties - Particle segregation and agglomeration - Sampling theory - Pre-treatments - Pre-processing methods - Performance-based modelling - Separation principles - Separation technologies - Ore - Tailings - Household waste - Waste incineration and Rankin cycle - Optimizing separation lines for value or recovery - Sensor feeding techniques - Sensor principles - Sensor applications - Detection-limited sensor data	
Course Contents Continuation		

(3) Recovery and quality of raw materials from waste
Link to learning objectives of the module:
B1: LO2, LO9, LO10 and LO12.

A challenging design and analysis exercise in student groups, spanning the route from waste source to extraction and further purification and/or refinement to high-quality raw materials.
Specific knowledge about mining operations, minerals processing, metallurgical refinement techniques, and final solutions for harmful residues (e.g., landfilling techniques) can be provided in a series of starting lectures or workshops, depending on the specified design topics. The following topics will be addressed:

Waste sources
Volumes and capacity
Conditions and limitations
Design of separation and treatment strategy and technologies
Quality of raw materials
Purification
Refinement
Treatments
Residues
Emissions
Energy consumption
Ethics
Ecological and societal impacts.

Quantification and discussion of environmental impacts in connection to regulations on residues and emissions are part of the evaluation of the technical design.

Moreover, students are challenged to identify and discuss possible ethical implications of their design in a dedicated workshop organised in collaboration with the Faculty of Technology, Policy, and Management.

Study Goals

On completion of this module, the students are able to:

B1-LO1: Recognise logistical procedures and key technologies that safeguard the recovery and quality of recyclable materials and reusable objects. (ILO: d, i)

B1-LO2: To take account of the main Dutch legislation and EU directives aimed at the circularity of materials from waste. (ILO: h, l)

B1-LO3: Make a considered choice for an efficient waste collection method. (ILO: c, d, l).

B1-LO4: Characterise solid waste heaps and streams (e.g. demolition waste/ore). (ILO: a, b, e)

B1-LO5: Match a pre-treatment to meet the requirements for optimum separation and quality of the raw materials. (ILO: a, d)

B1-LO6: Recognise appropriate design steps leading from the physical principle to an operational separation technology. (ILO: a, f, g)

B1-LO7: Make a considered choice for a separation technique or quality inspection technique. (ILO: a, g)

B1-LO8: Design and assess a waste/ore processing line in terms of recovery and quality or added value. (ILO: a, g)

B1-LO9: Implement efficient treatment steps that minimise solid residues and harmful emissions. (ILO: h, l)

B1-LO10: Quantify the non-circular impacts of a treatment option or process. (ILO: g, l)

B1-LO11: Take into account the main legislation concerning special wastes, residues, and emissions. (ILO: h)

B1-LO12: Design efficient treatment and processing steps that safeguard the recovery and quality of raw materials. (ILO: i, j, k)

Education Method

(1) Resources recovery from End-of-Life constructions

Physical lectures

Workshops followed by work in groups

Self-study

Students will be asked to prepare for the lectures/workshops, studying reading materials and/or scientific articles prepared for the lectures.

Following a thematic workshop, students will work in groups to produce a short report where, as an example, the pros and cons of a circular strategy are discussed, compared, and assessed quantitatively.

Background reading material is provided, which is sourced from research articles and book chapters from the scientific literature.

(2) Waste collection and separation techniques

Physical lectures

Online computer simulations in groups

Group assignments

Lab practical work in groups

(3) Recovery and quality of raw materials from waste

Physical lectures on special topics

Regular workshops and feedback sessions.

Assessment

(1) Resources recovery from End-of-Life constructions

Group assignment (15%): Group assignment: final report (10%) and oral presentation (5%). During the module, a peer review is conducted where each student submits feedback about his/her personal experiences on the level of collaboration and communication in the group.

(2) Waste collection and separation techniques

Group assignment: This assignment will involve experimental work in the laboratory (15%)

Written exam (20%)

(3) Recovery and quality of raw materials from waste

Group project (50%): final report (40%) and oral presentation (10%). During the module, a peer review is conducted where each student submits feedback about his/her personal experiences on the level of collaboration and communication in the group.

A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.

Expected prior Knowledge

-

Academic Skills

Being an interdisciplinary module, flexible and yet critical thinking, reflective functioning, and maintaining a collaborative and yet problem-solving attitude are the main targeted academic skills.

In addition, students will need to communicate and collaborate with teachers and technical assistants, which stimulates their

Literature & Study Materials	organizational and interpersonal skills. Hand-outs, articles/book chapters (via Brightspace), and open-source online material. (1) Resources recovery from End-of-Life constructions The European Green Deal, COM(2019) 640 final. A., Akhtar, A., K., Sarmah, Construction and demolition waste generation and properties of recycled aggregate concrete: A global perspective, Journal of Cleaner Production 186 (2018) 262-281 AT Gebremariam et al., Innovative technologies for recycling End-of-Life concrete waste in the built environment, Cons and Rec 2020, (163), 104911 Gálvez-Martos et al., Construction and demolition waste best management practice in Europe, Res, Cons and Rec 2018, (136), pp 166-178 (2) Waste collection and separation techniques Hand-outs, articles or book chapters via Brightspace and open-source material through the Internet. M.C.M. Bakker, Challenges in Advanced Solid Waste Separation, 2019, http://resolver.tudelft.nl/uuid:64a13fd4-edfe-4610-b20f-491278908281 EU Commission, Assessment of separate collection schemes in the 28 capitals of the EU, 070201/ENV/2014/691401/SFRA/A2, Final report 2015. G. Tchobanoglous, H. Theisen, S. Vigil, Integrated Solid Waste Management, McGraw-Hill. (3) Recovery and quality of raw materials from waste Hand-outs, scientific articles/books chapters (via Brightspace) and open-source online material; which will be tailored to the specific group topics.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	The B1-2 exam will be an open-book exam. No laptop will be permitted
Collegerama	No

ENVM1301	Reactive Resources and Wastes	15
Responsible Instructor	Dr. J. Gebert	
Instructor	Prof.dr.ir. M.K. de Kreuk	
Instructor	Dr. M.W.N. Buxton	
Instructor	Prof.dr.ir. J.B. van Lier	
Instructor	Prof.dr.ir. T.J. Heimovaara	
Instructor	Dr. J. Gebert	
Instructor	Dr.ir. R.E.F. Lindeboom	
Instructor	E. Maio	
Instructor	A. Soleymani	
Instructor	Dr. S.C. Buisma-Yi	
Contact Hours / Week x/x/x/x	x/x/x/10-16 (exact number differs over the weeks but is between 10 and 16)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This module addresses the processes in different types of reactive resources/wastes and their possible impact on the environment. These vary in relation to the different ways of material treatment, use or storage and disposal and will be evaluated for the different possible impact pathways (water path, gas path). On this basis, the potential of wastes to serve as secondary resources will be developed. Furthermore, this module will teach engineering applications to avoid and minimise environmental impacts during waste and resource storage, treatment, use and disposal, and to remediate and stabilise affected sites. The pertinent legislation will be taught as a continuous topic spanning all themes. The module content is organized in the following themes: (1) Environmental impact of reactive resources and wastes Link to learning objectives of the module: B2: LO1, LO2 and LO5. The following topics will be addressed: Water path: Leachate quality, transport of water through wastes and modelling thereof, leachate and groundwater monitoring, wastewater treatment technologies. Gas path: Gas production and modelling thereof; transport of gas through wastes and emissions into the atmosphere, gas concentration and emissions monitoring and treatment technologies. Greenhouse gas accounting and clean development mechanisms (CDM). Source-Pathway-Receptor-concept. Sorption, dispersion, transport. Point-of-compliance-point-of-emission concept. Acceptable emissions. Regulations on protection of soil, groundwater, surface water and atmosphere. (2) Waste as a resource Link to learning objectives of the module: B2: LO3 and LO5. The following topics will be addressed: Valorisation of wastes, recovery of raw materials from wastes. Concepts of waste-to-resource and waste-to-energy. End-of-waste criteria and procedures. Resource potential of specific waste streams: A- Biowastes: Anaerobic digestion B- Contaminated soils: Soil separation, treatment and remediation technologies (biological, chemical, physical). Re-use of soil. C- Sediments as a resource, sediments as a waste: Sediment-to-Soil ripening processes. Beneficial use of sediments. Sediment disposal. D- Mining residues: Properties, hazards, use potential. 'Hidden flows'. E- Construction and demolition wastes, bottom and other ashes. Regulations on re-use and recovery of wastes. (3) Final sinks Link to learning objectives of the module: B2: LO4 and LO5. The following topics will be addressed: Waste separation and treatment technologies Principles of sanitary landfill engineering including sealing, leachate collection and gas extraction systems Landfill processes and monitoring of emissions and deformation behaviour Landfill aftercare, end of aftercare and in situ waste stabilisation technologies Waste incineration Legacy sites Regulations concerning operation, monitoring and aftercare of sanitary and liability for legacy landfills.	
Study Goals	On completion of this module, the students are able to: B2-LO1: Explain the fundamental principles that guide contaminant mobilisation and emission from different types of waste. B2-LO2: Quantify contaminant emissions from waste via the water and gas/atmospheric pathway, analyse their impact on human health and the environment and conceptually design treatment options. B2-LO3: Apply end-of-waste criteria on potential secondary resources and analyse opportunities for and limitations to recovery, reuse and recycling of contaminated reactive materials. B2-LO4: Assess options and conceptually design develop stabilization and monitoring strategies for long-term safe use of legacy sites, sanitary landfills and mine tailing sites. B2-LO5: Understand current European waste-related legislation and apply the management options for reactive wastes as regulated therein.	
Education Method	Interactive lectures on theme-specific content Theme-specific self-study using scientific articles, reports or book chapters including presentation thereof to the class	

	Group project intersecting all three themes: case study As part of the case study, a workshop will be organized in collaboration with the Faculty of Technology, Policy and Management, in which students will discuss the ethical implications of the engineering solutions for the management of reactive resources and wastes. Field group work intersecting all three themes, including field measurements Excursion intersecting all three themes, e.g. to a waste sorting or treatment plant, wastewater treatment plant, landfill, incinerator, soil remediation project, dredged material treatment site.
Assessment	For the successful completion of the module, students have to pass the following assessments: Exam at module level (50%) Group project (30%) intersecting themes 1 to 3: technical report (25%) and final oral presentation (5%) Field group work, intersecting themes 1 to 3 (20%): report in the format of a scientific publication An intermediate self-reflection and fellow student peer feedback on performance of individual students to the group process is part of the group case study assignment.
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment. Contents taught in the preceding modules. High school level knowledge of chemistry and biological conversion processes. Basics of advective and diffusive flow.
Academic Skills	Reading of scientific literature, self-responsible and inquiry-based information collection and evaluation, planning and execution of a project, time management and anticipation, scientific writing and presenting, critical thinking and teamwork.
Literature & Study Materials	Textbook chapters, scientific papers, technical reports and lecture slides will be provided via Brightspace. Examples include: Christensen, T. (ed) (2010): Solid Waste Technology & Management, 1 & 2. Wiley, DOI:10.1002/9780470666883 Abdel-Mohsen & Paleologos (2017): Fundamentals of Geoenvironmental Engineering. Elsevier, ISBN: 9780128048306 Blume et al. (2016): Scheffer/Schachtschabel Soil Science. DOI:10.1007/978-3-642-30942 Hill, M. (2020): Understanding Environmental Pollution (4th ed.). Cambridge: Cambridge University Press. DOI:10.1017/9781108395021 Fitts, C. (2012): Groundwater Science. Academic Press, ISBN: 9780123847058 Appelo & Postma (2005): Geochemistry, Groundwater and Pollution, Second Edition. Taylor & Francis. ISBN: 978-0-415-36421-8 Hou, D. (2019): Sustainable Remediation of Contaminated Soil and Groundwater. Elsevier, ISBN: 9780128179826.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Open book, all materials except electronic devices which are connected to the internet.
Collegerama	No

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